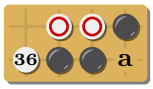
# pgf-go




**Manual for version 0.2026.10.14**

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# 1 Introduction

## 1.1 Motivation

The following package offers a collection of macros designed to represent Go boards and positions, fully implemented in PGF. This project, although still under development, incorporates innovative features aimed at simplifying and streamlining the creation of Go-related diagrams:

- Profile management for boards and players.
- Loading multiple coordinates simultaneously using a syntax that some might find familiar.

The development of PGFGO stems from frustration with not finding packages that met my needs. The packages I explored allow for the representation of black-and-white boards, suitable for printing, but impractical when combined with technologies like `beamer` or in documents where a color presentation is desired.

Initially, PGFGO was intended to be limited to these tasks; however, I now envision it more as an interface for managing styles and states, avoiding excessive use of direct formatting. This will be reflected primarily in the structure of the *keys* tree.

Regarding the name, although the macros are implemented in PGF and L3, the board is built on `\tikzpicture` so that users familiar with this environment can take advantage of its macros.

## 1.2 What's New

- 0.2026.10.14:
  - The computations (coordinates, move numbers, memory, etc.) have been separated from the drawing. They are done by a *kernel*, chosen with the package option `kernel` (p. 12). Besides the kernel written in L3, a kernel written in Lua has been added, which is the default with LuaLaTeX.
  - With the Lua kernel, captures (`captures`, p. 41) and SGF games (`\sgf`, p. 53) have been added. With the `animate` package, `\sgfanimation` steps through a game with buttons.
  - "N at M" notes have been added for numbered moves on points already played (`notes` and `\pgfgomovenotes`, p. 45).
  - SGF: variations (`variation`), the marks of the file (`marks`), the game information (`\sgfinfo`), numbering with `MN` and `restart`, and the board size taken from the file (`sgf size`).
  - SGF: collections of games (`game`), the view (`VW`, `sgf view`), figures (`FG`) in `\sgfdiagrams`, variations as letters (`variations`), `PL` and Latin-1 files.
  - SGF: node names and annotations (`TE`, `BM`, `GB`, etc.) are written with the comments. `\sgfcomment` writes the comment of a move, and `\sgfdiagrams` can write the comments below each diagram (`comments`).
  - Arrows and lines (`\arrow` and `\line`, p. 68) and the `dim` mark have been added. `\sgf` also draws `AR`, `LN`, `SL` and `DD`.
  - The board draws the star points (`hoshi`, p. 33).
  - `\territory`, which marks the territory (with a polygon, or by itself with `auto` and the Lua kernel), and `\pgfgoscore`, which counts the points, have been added.
  - `\sgfanimation` builds the animation by layers (`layers`): every stone is drawn only once, and the PDF of a long game is much smaller.
  - With `captured marks = remove`, the marks on a captured stone go with it.
  - `\pgfgoprisoners`, which counts the prisoners of each player, and `\sgfdiagrams`, which draws a game as a series of diagrams, as books do, have been added.
- 0.2025.10.13: Bug fixes. Each `goban` starts at move 1 with Black, `\mark` and `\marks` only exist inside `goban`, and what falls outside a partial board is no longer drawn. The full list is in the file `CHANGELOG.md`.
- 0.2025.03.10:
  - The *keys* `inner` and `outer` have been added for the *rounded corner* of the board.
  - Three new coordinate operators have been added: `*` for all intersections, `L` for vertical lines, and `N` for horizontal lines.
- 0.2025.03.07: First release.

# 2 The Boards of Kāru

## 2.1 Board and Stones

Kāru is writing an article to share with his students. He has noticed that diagramming *software* does not offer satisfactory formats or results. Therefore, he decides to try PGFGO and starts by representing a game like the one shown in Figure 1.

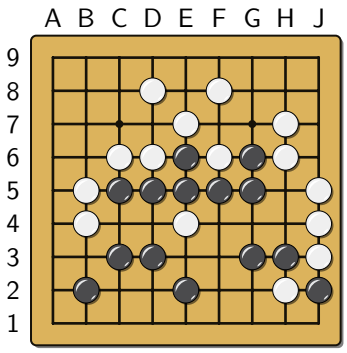
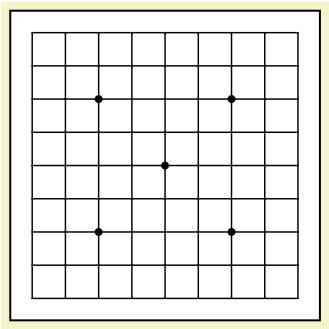


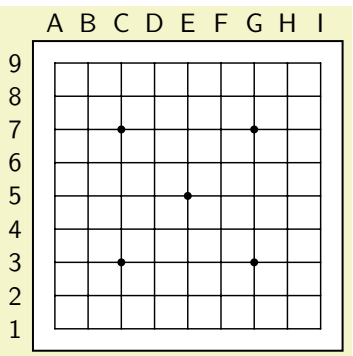

Figure 1

Kāru must start with the board. In PGFGO, there is currently a single environment where elements can be placed: `goban`. This environment controls by default the size and color of the board, the position and type of labels, among other properties. It also allows partial board representations.



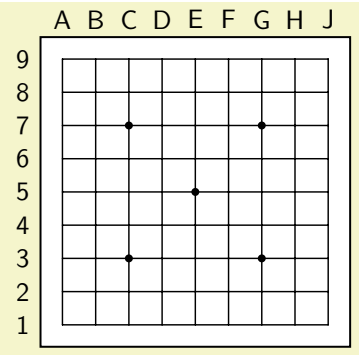
```
\begin{goban}
\end{goban}
```

Kāru is satisfied. The environment provides a default $9 \times 9$ board, which is exactly what he needs, but he must add side labels. The `goban` environment does not require mandatory parameters, although it accepts an optional one: a list of `keys` or `key=values` for configuration. Kāru consults the section on `goban` (p. 30) and decides to add labels to the environment. He knows that the *keys* controlling the objects in the goban are divided into three groups: `grid`, for the central grid; `background`, for the border and board fill; and `label`, for coordinate labels. The latter is the one that interests him. This *key*, among other options, can take the *key* `at` to define the label position with values such as `above`, `west`, `below`, `right`, `all`, or `none`. Cardinal points can also be used. Since he only wants labels above and to the left of the goban, he will use the *keys* `above` and `left`.




```
\begin{goban}[label = {at = {above, left}}]
\end{goban}
```

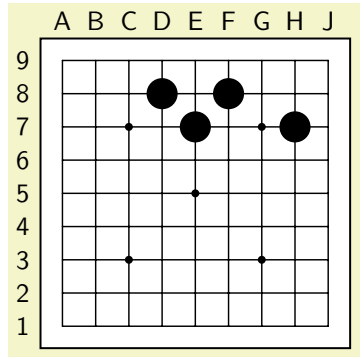
Kāru feels that everything is going well, but he notices that his board includes an `I` instead of omitting it and going directly to `J`. By default, in PGFGO, "I" exists and is used both as a coordinate and in the side labels, although it is possible to remove it. There are two ways to do this: setting the global *key* `use i` to `false` (`\pgfgoset{use i = false}`), which would prevent Kāru from worrying about this throughout the rest of the article, or applying it directly to the board with the same *key* and value.




```
\pgfgoset{use i = false}

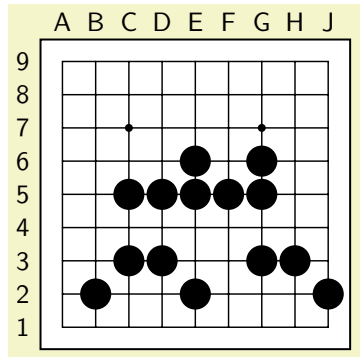
\begin{goban}[label = {at = {above, left}}]
\end{goban}
```

Now he is ready to place the stones. For this, Kāru has several macros. Let's start with the most basic one: `\stone[options]{intersection}`. This command accepts two parameters: the first, optional, controls the appearance and additional elements of the stones (such as marks); the second, mandatory, is the intersection. Kāru does not need to convert intersections to numbers since he can specify them alphanumerically. By default, the stones are neutral, but he can choose the first player by passing the *key* `black` or `b`.




```
\begin{goban}[label = {at = {above, left}}, use i = false]
 \stone[black] {D8}
 \stone[black] {F8}
 \stone[b] {E7}
 \stone[b] {H7}
\end{goban}
```

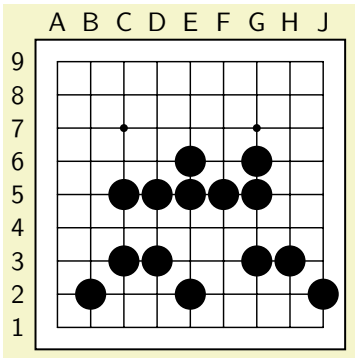
Kāru suspects that this is not the best option if he needs to add 200 stones or more. To simplify the syntax, he finds in the documentation the macro `\stones[options]{intersection, intersection, ...}`, which accepts the same optional *keys* as `\stone`.[1]




```
\begin{goban}[label = {at = {north, west}}, use i = false]
 \stones[b]{B2,E2,J2,C3,D3,G3,H3,C5,D5,E5,F5,G5,E6,G6}
\end{goban}
```

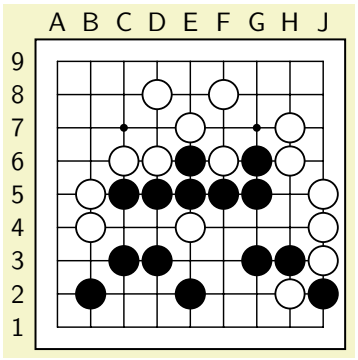
Kāru also notices that there is a syntax (p. 12) to add several coordinates at once. For example, for a line of stones from C5 to G5, he can write `C:G5` or `C5--G5`.

[1] Between `\stone{A1}` and `\stones{A1}`, the first one is much faster. We will see later that many processes are involved in reading the mandatory argument of the plural macros.



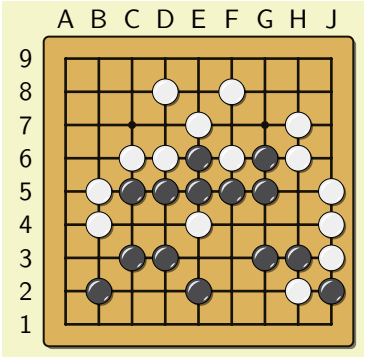
```
\begin{goban}[label = {at = {north, west}}, use i = false]
 \stones[b]{B2,E2,J2,C3,D3,G3,H3,E6,G6,C:G5}
\end{goban}
```

To add white stones, Kāru must include the *key* `white` in the `\stones` command. He knows that `white` and `black` do not literally refer to the colors white and black, but to the second and first player, respectively, and that when these *keys* are applied, the objects inherit their properties.



```
\begin{goban}[label = {at = {north, west}}, use i = false]
 \stones[b]{B2,E2,J2,C3,D3,G3,H3,E6,G6,C:G5}
 \stones[w]{D8,F8,E7,H7,C6,D6,F6,H6,B5,B4,E4,H2,J3:5}
\end{goban}
```

Finally, the style of his board remains. For this, he looks through the catalog (p. 72) and finds a style he likes: `pgfgo`. To apply it, he reads the section on profiles (p. 19) and decides to use the macro `\pgfgouseprofile{<name>}`.



```
\pgfgouseprofile{pgfgo}

\begin{goban}[label = {at = {north, west}}, use i = false]
 \stones[b]{B2,E2,J2,C3,D3,G3,H3,E6,G6,C:G5}
 \stones[w]{D8,F8,E7,H7,C6,D6,F6,H6,B5,B4,E4,H2,J3:5}
\end{goban}
```

## 2.2 Moves

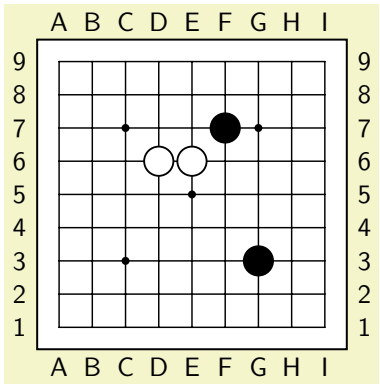
When the stones are to be placed in order, following the sequence ”black, white, black, white”, it can be tedious and unnecessary to give the *key* of each player.

```
\begin{goban}
 \stone[b] {F7}
 \stone[w] {E6}
 \stone[b] {G3}
 \stone[w] {D6}
\end{goban}
```

That is why PGFGO includes macros that make this easier. Let’s start with the singular one:

`\move[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

This command works like `\stone`, but alternates the colors automatically.

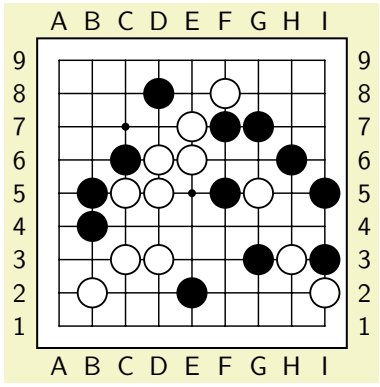


```
\begin{goban}[label = {at = all}]
 \move {F7}
 \move {E6}
 \move {G3}
 \move {D6}
\end{goban}
```

The macro `\move`, like `\stone`, has a plural version:

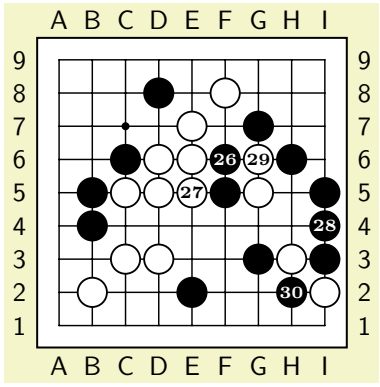
`\moves[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

In this case, giving the *key* `black` or `white` only sets who is the first player of the sequence.



```
\begin{goban}[label = {at = all}]
 \stone[b] {F7}
 \moves[white] { E6,G3,D6,F5,E7,E2,F8,E6,D8,D3,
  G7,C5,C6,D5,B5,G5,B4,B2,H6,C3,
  I3,I2,I5,H3}
\end{goban}
```

The macros `\move` and `\moves` also have "starred" versions (`\move*` and `\moves*`), which label the moves. The *key* `from` sets the number where the labels start.



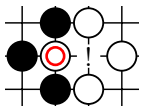

```
\begin{goban}[label = {at = all}]
 \moves[white] { E6,G3,D6,F5,E7,E2,F8,E6,D8,D3,
  G7,C5,C6,D5,B5,G5,B4,B2,H6,C3,
  I3,I2,I5,H3}

 \moves*[from = 26]{F6,E5,I4,G6}
 \move* {H2} % Black and labeled 30, without saying so.
\end{goban}
```

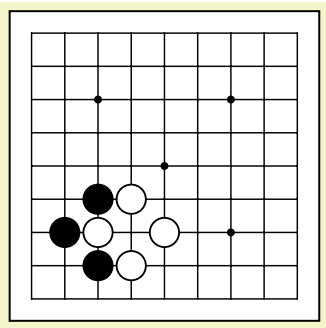
In this last example, `H2` was placed with `\move*` and, although it is separate from `\moves*`, it is black and labeled 30, because the last stone placed (`G6`) was black and labeled 29.

## 2.3 Partial Board and Marks

Sometimes we want to focus on a particular region of the board. For this reason, it is possible to show only a portion of it.

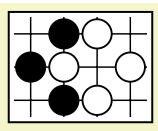


To draw this board, we start by placing the stones.



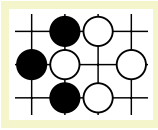
```
\begin{goban}
 \stones[b] {C2,B3,C4}
 \stones[w] {D4,E3,D2,C3}
\end{goban}
```

With the *key* `partial`, we can set the range of the board through its *key* `from`. It takes a value of the form `LN to LN`, where each `LN` is one of a pair of opposite vertices of the rectangle.



```
\begin{goban}[partial = {from = B2 to E4}]
 \stones[b] {C2,B3,C4}
 \stones[w] {D4,E3,D2,C3}
\end{goban}
```

To remove the line of the board, we use the *key* `line` of the `background` of the `goban`.



```
\begin{goban}[partial = {from = B2 to E4}, background = {line = false}]
 \stones[b] {C2,B3,C4}
 \stones[w] {D4,E3,D2,C3}
\end{goban}
```

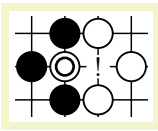
Now, to place marks we have the macro

`\mark[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

and its plural version.

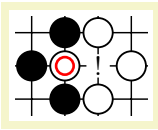
This macro is quite broad and is documented on its own page. For now, we need to know two things:

- The mark for the circle is called `circle`.
- The mark "!" can be added with `custom`.



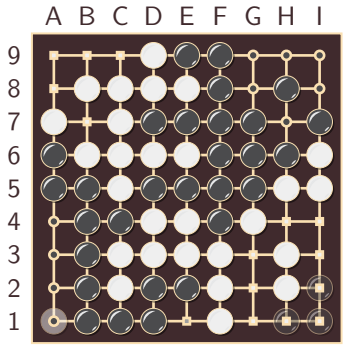
```
\begin{goban}[partial = {from = B2 to E4}, background = {line = false}]
 \stones[b] {C2,B3,C4}
 \stones[w] {D4,E3,D2}
 \mark[custom = {!}] {D3}
 \stone[w, mark = circle] {C3}
\end{goban}
```

We can change the line color of a mark with the *key* `mark/line/color`.

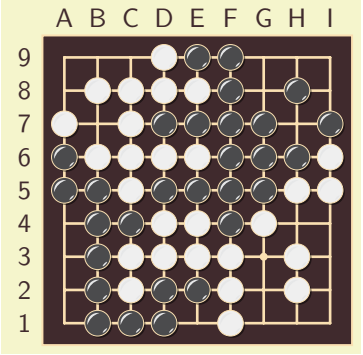


```
\begin{goban}[partial = {from = B2 to E4}, background = {line = false}]
 \stones[b] {C2,B3,C4}
 \stones[w] {D4,E3,D2}
 \mark[custom = {!}] {D3}
 \stone[w, mark = {circle, line = {color = red}}] {C3}
\end{goban}
```

## 2.4 The End of the Game: Territory




Let's start from the following board:



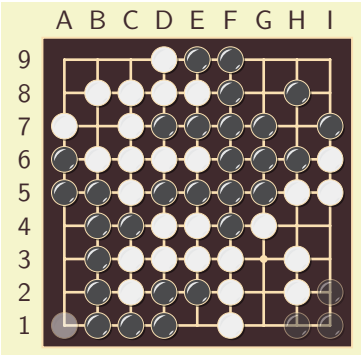

```
\pgfgouseprofile{wain}

\begin{goban}[label = {at = {above, left}}]
 \stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
 \stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}
\end{goban}
```

The next step is to identify the prisoner stones. Prisoners can be marked with the *key* `prisoner` or with the macro

`\prisoner[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

which also has a plural version.

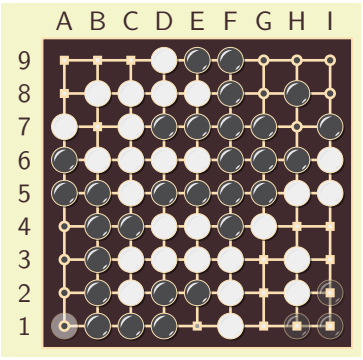



```
\pgfgouseprofile{wain}

\begin{goban}[label = {at = {above, left}}]
 \stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
 \stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}

 \stones[b, prisoner]{H1-|I2} % Same as \prisoners[b] {...}
 \prisoner[w] {A1} % Same as \stone[w, prisoner] {...}
\end{goban}
```

Now let's define the territories. This can be done in many ways. In particular, two are implemented for sure: the mark `t-square` and the mark `t-circle`. These marks always take as fill the fill color of the stone they belong to, and as border color and width those of the grid.



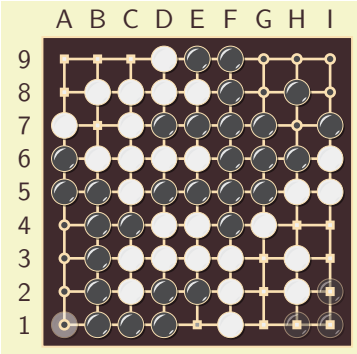

```
\pgfgouseprofile{wain}

\begin{goban}[label = {at = {above, left}}]
 \stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
 \stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}

 \stones[b, prisoner]{H1-|I2}
 \prisoner[w] {A1}

 \marks[b, t-circle]{A1:4, G8|-I9, I8, H7}
 \marks[w, t-square]{G3|-I1|-H4, C9-|A8, B7}
 \marks[t-square, fill = {color = red}] {E1} % Not filled in red: it takes the neutral color.
\end{goban}
```

Another way is `\territory` (p. 42). With a polygon, it marks the points of that region that have no living stone; with `auto` and the Lua kernel, it finds the territory of each player by itself. `\pgfgoscore` writes the score.



Black: 11. White: 23.5.

```
\pgfgouseprofile{wain}

\begin{goban}[label = {at = {above, left}}]
 \stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
 \stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}

 \stones[b, prisoner]{H1-|I2}
 \prisoner[w] {A1}

 \territory[dame]{auto}
\end{goban}

Black: \pgfgoscore{black}. White: \pgfgoscore[komi = 6.5]{white}.
```

# 3 The keystone

## 3.1 The `\let` Game

`\mark` and `\marks` are TEX primitives. To leave them untouched, the PGFGO macros `\mark` and `\marks` only exist inside the `goban` environment. Outside it they are still the primitives, so page marks and running heads are not affected. `\sgf` does not exist outside `goban` either.

## 3.2 Configuration

The macro `\pgfgoset{...}` allows defining general configurations for the package. It is equivalent to `\keys_set:nn{pgfgo}{...}`, meaning it uses the LaTeX3 key system to set options in a structured and hierarchical manner. This macro is particularly useful for customizing the package's global behavior, such as the appearance of the board, stones, or moves, without needing to repeat configurations in each individual environment. For example, a user could define custom colors for players or adjust the board's design to apply to all diagrams in the document. It also serves as a gateway to direct formatting.[2]

`\pgfgoset{`⟨*options*⟩`}`

`/pgfgo/goban` = ⟨*goban options*⟩ (no default)

Sets general options for the board. This includes properties such as size, background color, visibility of grid lines, or coordinate labels. For example, it can be used to create partial boards or modify the visual style of the border.

`/pgfgo/stone` = ⟨*stone options*⟩ (no default)

Defines general options for stones, such as their fill color, outline, or default markings. It is useful for establishing a consistent style across all stones in a document (neutral, white, or black).

`/pgfgo/move` = ⟨*move options*⟩ (no default)

Sets general options for moves, such as the format of their numbers. It is the same as `\pgfgoset{players = {all = {m`

`/pgfgo/use i` =⟨*boolean*⟩ (default `true`)

Allows deciding whether or not to use the "I" label on the board. By default, it is enabled ("true"), meaning coordinates include the letter "I". If disabled ("false"), the system skips this letter and jumps directly from "H" to "J", which is common in certain Go game representations.

`/pgfgo/sgf animation` = ⟨*sgf animation options*⟩ (no default)

Options of every `\sgfanimation` in the document (p. 53).

`/pgfgo/sgf diagrams` = ⟨*sgf diagrams options*⟩ (no default)

Options of every `\sgfdiagrams` in the document (p. 53).

`/pgfgo/sgf annotations` = ⟨*texts*⟩ (no default)

The texts of the annotations of SGF files (p. 53).

`/pgfgo/arrow` = ⟨*arrow options*⟩ (no default)

Options of every arrow in the document (p. 68).

`/pgfgo/line` = ⟨*line options*⟩ (no default)

Options of every line in the document (p. 68).

`/pgfgo/territory` = ⟨*territory options*⟩ (no default)

Options of every `\territory` in the document, like `draw = fill` (p. 42).

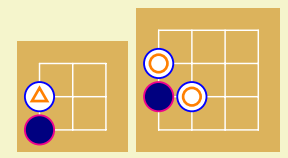
`/pgfgo/score` = ⟨*score options*⟩ (no default)

Options of every `\pgfgoscore` in the document, like `komi = 6.5` (p. 42).

[2] Use with discretion…unlike footnotes.

/pgfgo/players = ⟨*players options*⟩ (no default)

Allows defining options for each player (including neutral ones). This *key* acts as a container encompassing specific configurations for black, white, and neutral players, facilitating customization based on the user's rules or preferences. All available options and how to customize stone styles will be detailed in another section.



```
\pgfgoset{
 goban = {
  background = {
   fill = {color = pgfgo},
   line = false},
  grid = {line = {color = white}}},
 players = {
  black = {
   stone = {
    line = {color = magenta},
    fill = {color = blue!50!black}}},
  white = {
   stone = {line = {color = blue}},
   mark = {line = {color = orange}}
  }
 }
}

\begin{goban}[3]
 \stone[black] {A1}
 \stone[white, mark = {triangle}] {A2}
\end{goban}

\begin{goban}[4]
 \stone[black] {A2}
 \stones[white, mark = {circle}] {B2,A3}
\end{goban}
```

Subsequent `goban` environments inherit these configurations, although they can be overridden locally if necessary.

## 3.3 The Compute Kernel

PGFGO separates computing from drawing. The computations (reading coordinates, knowing which part of the board is visible, numbering moves and keeping track of the turn, remembering boards, etc.) are done by a *kernel*. The drawing, the *keys* and the profiles are always the same. There are two kernels:

- `lua`: written in Lua. It needs LuaLaTeX, and it is the default with LuaLaTeX.
- `expl3`: written in L3. It works with any engine, and it is the default with the other ones.

Both produce exactly the same diagrams. The Lua kernel can also do three things that the other one cannot: captures (p. 41), finding the territory with `\territory{auto}` (p. 42) and reading SGF games (p. 53). This manual is compiled with the Lua kernel.

/pgfgo/package/kernel = ⟨*expl3 or lua*⟩ (no default, initially `lua with LuaLaTeX, expl3 with other engines`)

Package option that chooses the kernel. It is only needed to use `expl3` with LuaLaTeX. It can only be given when PGFGO is loaded; it cannot be changed with `\pgfgoset`. If `kernel = lua` is requested with pdfLaTeX, the package warns and uses `expl3`. With `expl3`, `captures`, `\territory{auto}` and `\sgf` raise an error.

```
\usepackage[kernel = expl3]{pgf-go}
```

## 3.4 Coordinate Syntax

One feature that distinguishes PGFGO from other packages is the ability to input coordinates using various syntaxes, offering flexibility to represent positions on the board intuitively.

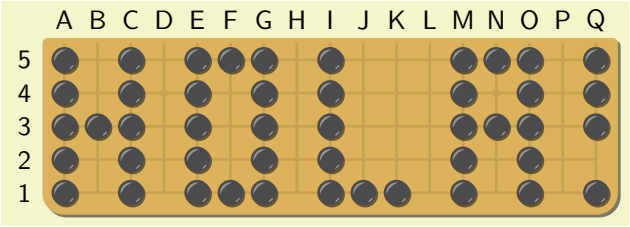
- Multiple entries can be separated by commas.
- Stones can be removed by adding a subtractive group after a semicolon («;»).

However, for performance and consistency reasons, some freedoms that were initially unrestricted have been limited. For example, to refer to an intersection, the alphabetical coordinate must be written in uppercase and without spaces. Thus, the B5 intersection must be entered as `B5`, not as `B 5`, `B  5`, or `b5`, since spaces between the letter and number are invalid in this context. Restrictions will be explained in each case.

When PGFGO has to read the coordinate argument, it may encounter something like this:

```
\stones{<item>, <item>, …; <item>, <item>, …; …}
```

The first thing it will try to do is split the argument at the `;` and build two groups: the additive group and the subtractive group. The additive group is responsible for adding stones using our syntax. The subtractive group is responsible for removing them. These groups alternate automatically when a `;` is placed. This is explained in more detail in the corresponding subsection. If no `;` is placed, PGFGO will add all the stones to the additive group. Multiple stones or multiple groups separated by `,` can be added. Together, all this provides an interface to create complex groups with few *tokens*:



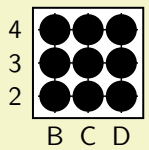

```
\pgfgouseprofile{pgfgosmooth}

\begin{goban}[17, partial = {from = A1 to Q5}, label = {at = {north, west}}]
 \stones[b]{*;B,D,H,L,Q2,P,F2:4,J2rK5,N1:2,N4;B3}
\end{goban}
```

### 3.4.1 Stone Groups

**All the stones on the board (*Why not?*)**

Adds all the **visible** intersections of the board to the corresponding group. That is, if it is a partial board, only those corresponding to the visible region will be added. It is important to clarify this in case it is used in combination with `remember`. Its syntax is `*`.

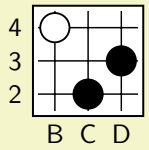



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[label = {at = {south, west}}]
 \stones[b]{*}
\end{goban}
```

**Intersection**

Intersections are specified by indicating the letter followed by the number, with no spaces between the alphabetic and numeric components.




```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

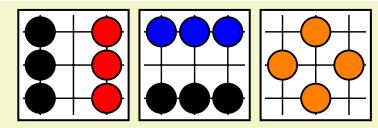
\begin{goban}[label = {at = {south, west}}]
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \stone[w]{B 3} % Not recognized because it has a space.
\end{goban}
```

In the example, `B 3` is not recognized due to the space, while `B4` is correctly drawn as a white stone.

**Vertical or Horizontal Lines**

Lines are groups of vertical or horizontal stones implicitly delimited by the size of the board.

- For horizontal lines, the syntax is `L`.
- For vertical lines, the syntax is `N`.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{B2,B3,B4}
 \stones[fill = {color = red}]{D}
\end{goban}

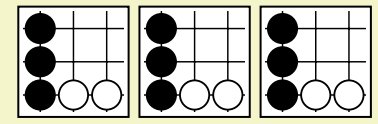
\begin{goban}
 \stones[b]{B2,C2,D2}
 \stones[fill = {color = blue}]{4}
\end{goban}

\begin{goban}
 % Adding lines C and 3. Then removing C3.
 \stones[fill = {color = orange}]{C, 3; C3} % NOTE: C,3 != C3.
\end{goban}
```

**Segments**

Segments are explicitly delimited groups of vertical or horizontal stones. Since they are unidirectional (either vertical or horizontal), one of their coordinates remains constant and does not need to be repeated; only the range of the other variable is specified.

- For horizontal segments, the syntax is `L:LN` or `L:L N`.
- For vertical segments, the syntax is `LN:N` or `L N:N`.



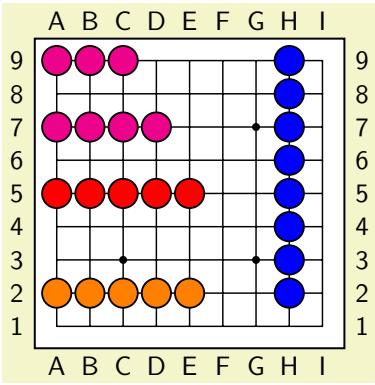
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{B2,B3,B4}
 \stones[w]{C2,D2}
\end{goban}

\begin{goban}
 \stones[b]{B2:4}
 \stones[w]{C:D2}
\end{goban}

\begin{goban}
 \stones[b]{B 2:4} % Spaces work between distinct elements.
 \stones[w]{C:D 2}
\end{goban}
```

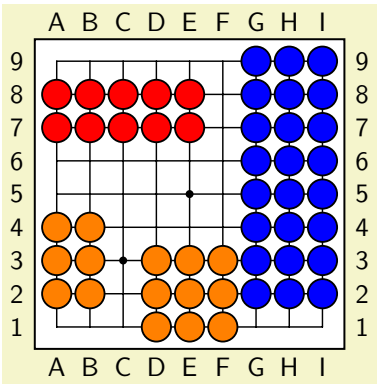
Here, `B2:4` represents a horizontal segment from B2 to B4, while `C:D2` indicates a horizontal segment from C2 to D2.




```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = red}]{A:E5}
 \stones[fill = {color = blue}]{H9:2}
 \stones[fill = {color = orange}]{E:A2}
 \stones[fill = {color = magenta}]{A:D7, A:C9} % Can be combined.
\end{goban}
```

**Rectangle: An Artificial Consequence**

The segment syntax allows creating ranges between coordinates to insert stones. I found it useful to extend it to include two ranges, offering a first approximation to rectangles (though it's not my favorite method, here it is). In this case, the syntax is `L:L N:N` or `L:LN:N`.




```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = red}]{A:E 7:8} % Space is not mandatory.
 \stones[fill = {color = blue}]{I:G 9:2}
 \stones[fill = {color = orange}]{A:B 2:4, F:D 1:3}
\end{goban}
```

For example, `A:E 7:8` creates a rectangle spanning columns A to E in rows 7 and 8, while `I:G 9:2` descends from I9 to G2. Although functional, this syntax can be less intuitive, leading to more specific operators we'll see later.

### Cartesian Products

In this context, the Cartesian product is a more powerful extension of segments and, at the same time, a generalization of all the notations presented so far. The idea is the following: suppose we want to enter a group of stones on column A: `A1:3, A5, A7:8`. Until now there was no simpler way to write this. But every item repeats the A at the beginning: what if, instead of writing it three times, we write it only once? Something like

`A[1:3, 5, 7:8]`

In this case, the interpreter distributes the A on the left to each of the elements on the right.

The elements of the list on the right do not have to be numbers. Something like `A:D 5:7, A1:2, A3, A:G9` can also be rewritten as

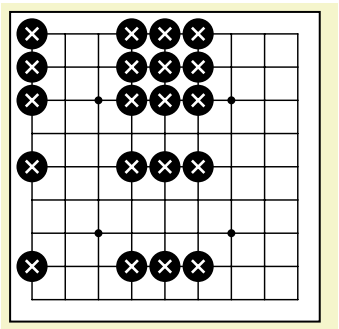
`A[:D 5:7, 1:2, 3, :G9]`

Of course, nobody abstracts that much when entering coordinates, so this is a mere whim. The only restriction on the content between brackets is that brackets cannot be nested.

Just as there is a syntax for letters on the left, there is one for numbers on the right, and it works the same way. The expression `A12, C12, E:G12` can be written as `[A,C,E:G]12`.

There is a third operator, which, yes, is more of a whim than anything else: the unrestricted Cartesian product. I will not dwell on it, because it requires a higher degree of abstraction and has no real human use. This operator is a combination of the previous two: the expression `[A,D:F][2,5,7:9]` is read as

`A2, A5, A7:9, D:F2, D:F5, D:F 7:9.`



```
\begin{goban}
 \stones[b]{[A, D:F][2, 5, 7:9]}
 \marks[b]{A2, A5, A7:9, D:F2, D:F5, D:F 7:9} %<- what the interpreter reads.
\end{goban}
```

The Cartesian product syntax can therefore be used in three different ways:

- `L[itm1, itm2,...]`.
- `[itm1, itm2,...]N`.
- `[itm1,itm2,...][itm3,itm4,...]`.

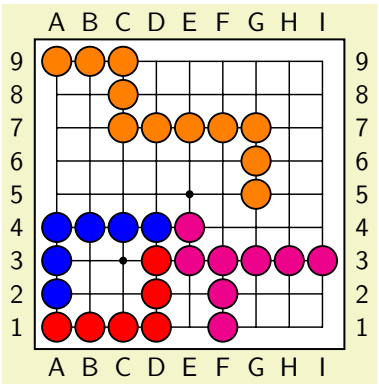
where brackets cannot be nested.

It can also be combined with some of the operators we will see next to create nodes along paths.

So far, we've covered commands for intersections that cannot be combined with others in the same instruction and therefore **must** be separated by commas. Now, we'll explore commands that can be chained together, offering greater flexibility for complex patterns.

**Horizontal–Vertical and Vertical–Horizontal**

Given two intersections that are neither vertically nor horizontally aligned, these commands generate the intersections connecting the first to the second via a "horizontal–vertical" or "vertical–horizontal" path. This may remind you of the TikZ `-|` operator, from which I borrowed the idea. The syntax is `LN -| LN` (for the first case) or `LN |- LN` (for the second), with or without spaces. These commands, along with the next one, can be chained to form ladders or paths typical of the Manhattan metric.

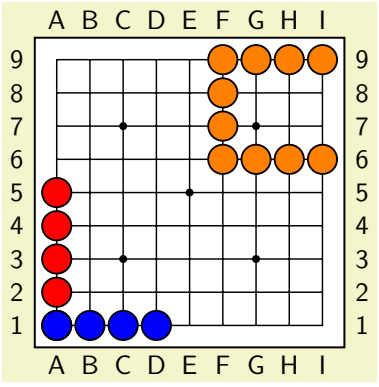


```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = red}]{A1 -| D3}
 \stones[fill = {color = blue}]{D4 -| A2}
 \stones[fill = {color = orange}] {A9 -| C7 -| G5} % Chained.
 \stones[fill = {color = magenta}] {F1 |- I3 -
| E4} % Try to guess.
\end{goban}
```

In this example, `A1 -| D3` goes horizontally from A1 to D1 and then vertically to D3.

**Horizontal or Vertical Line**

Given two intersections aligned vertically or horizontally, this operator connects them with a straight line of stones. Its syntax is `LN -- LN`. It can be chained with the previous two commands to form more elaborate patterns. If the intersections are not aligned, the operator draws nothing (who knows, maybe in the future it will generate ladders ;)).



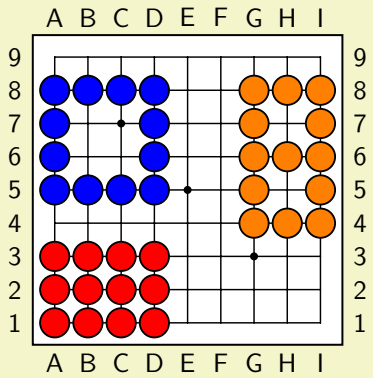
```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = red}]{A1 -- A5}
 \stones[fill = {color = blue}] {A1 -- D1}
 \stones[fill = {color = magenta}] {A6 -- B9} % Doesn't draw because they're not al
 \stones[fill = {color = orange}] {I9 -| F6 -- I6} % Chained.
\end{goban}
```

Here, `A1 -- A5` draws a vertical line, while `I9 -| F6 -- I6` combines a "horizontal–vertical" path with an additional horizontal line.

Beware! `A1 -- D1 -| F3` is a permitted chain, while `A:D1 -| F3` is not, because `:` is not a chain operator. This might change in future versions of the package.

**Filled and Empty Rectangles**

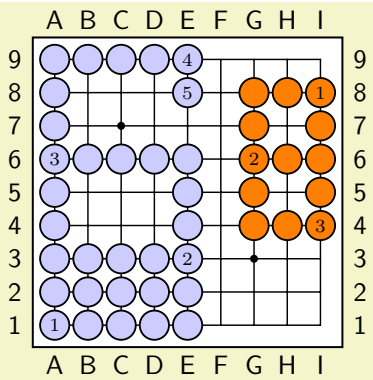
The previous rectangle syntax `L:L N:N` is unintuitive and more of a legacy of bounded segments. Therefore, there are two more direct operators for creating rectangles: `R` and `r`. The difference is that `R` generates a filled rectangle, while `r` only draws its outline. Given two diametrically opposite vertices of the rectangle, the syntax is `LN R LN` or `LN r LN`. Although spaces are not currently mandatory, I recommend using them for better readability.



```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = red}]{A1 R D3}
 \stones[fill = {color = blue}] {A5 r D8}
 \stones[fill = {color = orange}] {I8 r G4, H6} % It's alive!
\end{goban}
```

For example, `A1 R D3` fills a rectangle from A1 to D3, while `A5 r D8` draws only the border of a larger rectangle.

Rectangles are also chain operators and can be used as such to create more complex patterns.

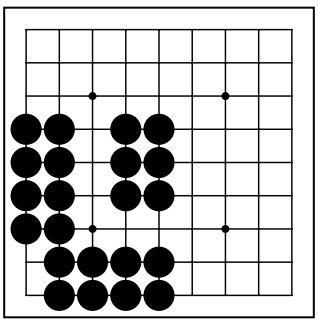


```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = orange}] {I8 r G6 r I4} % You guess.
 \marks[sequence]{I8,G6,I4}
 \stones[fill = {color = blue!20}] {A1 R E3 r A6 |- E9 -- E8} % Wow!
 \marks[sequence]{A1,E3,A6,E9,E8}
\end{goban}
```

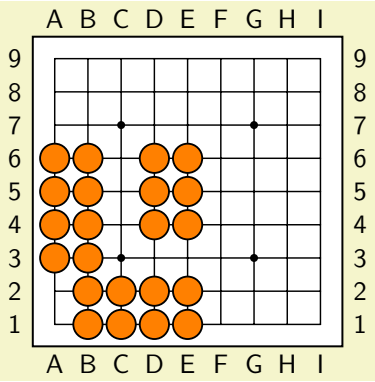
In this case, `I8 r G6 r I4` chains two outline rectangles, while `A1 R E3 r A6 |- E9 -- E8` combines a filled rectangle, an outline, a "vertical–horizontal" path, and a line, showcasing the power of chaining.

### 3.4.2 Additive and Subtractive Groups

Suppose we want to draw the following diagram:

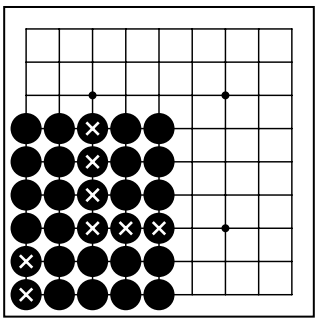


This can be achieved with the operators already seen, breaking the pattern into several parts:

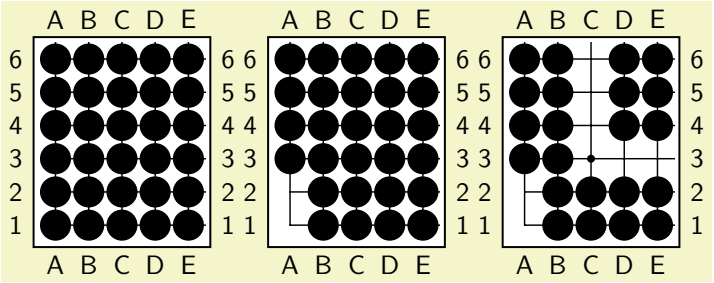


```
\begin{goban}[label = {at = all}]
 \stones[fill = {color = orange}] {A6 R B3, B2 R E1, D4 R E6}
\end{goban}
```

But we could also imagine starting from a complete rectangle from which we've removed the marked stones, offering a more conceptual perspective:



This is what additive and subtractive groups are about. An additive group adds intersections to the list, while a subtractive group removes them. By default, commands always begin processing additive groups, which explains why they draw stones directly. In PGFGO, we can switch between both types using `;`.



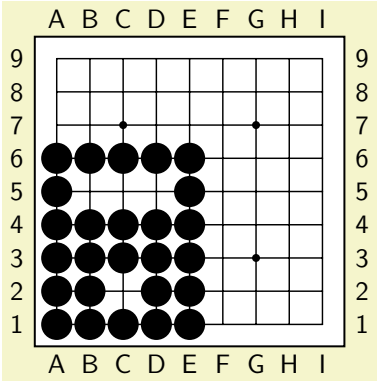
```
\begin{goban}[label = {at = all}, partial = {from = A1 to E6}]
 \stones[b]{A1 R E6} % Additive group.
\end{goban}

\begin{goban}[label = {at = all}, partial = {from = A1 to E6}]
 \stones[b]{A1 R E6; A 1:2} % Additive group; subtractive group.
\end{goban}

\begin{goban}[label = {at = all}, partial = {from = A1 to E6}]
 \stones[b]{A1 R E6; A 1:2, C6 |- E3} % Additive group; subtractive group.
\end{goban}
```

In this sequence, a full rectangle is drawn first (`A1 R E6`), then the horizontal segment `A1:2` is removed, and finally, the path `C6 |- E3` is subtracted. This results in the same diagram as the initial example but with a more compact approach.

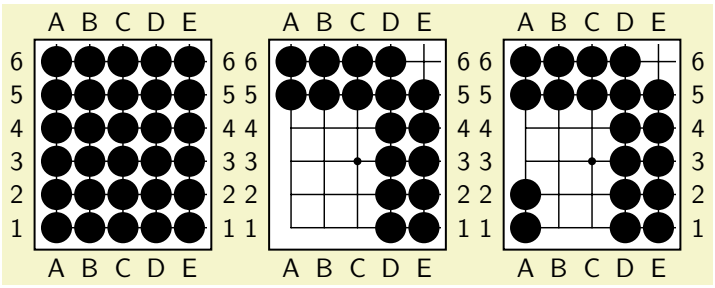
The advantage of this system is that it allows adjustments when needed. Sometimes, an additive pattern is ideal except for one or two stones; in those cases, we can remove them with a subtractive group without redesigning everything.



```
\begin{goban}[label = {at = all}]
 \stones[b]{A1 R E6; B:D 5, C2}
\end{goban}
```

Here, `B:D 5` removes a horizontal segment in row 5, and `C2` removes a specific stone, adjusting the initial rectangle.

Note that additive and subtractive groups can be chained, separated by `;`, allowing alternation between adding and removing elements in a single line as many times as desired.



```
\begin{goban}[label = {at = all}, partial = {from = A1 to E6}]
 \stones[b]{A1 R E6} % AD
\end{goban}

\begin{goban}[label = {at = all}, partial = {from = A1 to E6}]
 \stones[b]{A1 R E6; A1 R C4, E6} % AD, SU
\end{goban}

\begin{goban}[label = {at = all}, partial = {from = A1 to E6}]
 \stones[b]{A1 R E6; A1 R C4, E6; A 1:2} % AD, SU, AD
\end{goban}
```

In this last example, a rectangle is created, two regions are removed (`A1 R C4` and `E6`), and then a segment is added (`A 1:2`).

## 3.5 Profiles

Profiles are equivalent to styles in Ti*k*Z. In short, they allow saving a board and stone configuration under a name and recalling it whenever desired.

`\pgfgodefinegoban[`⟨*options*⟩`]{`⟨*goban name*⟩`}{`⟨*goban options*⟩`}`

Allows assigning a name to a group of goban options. It has two optional settings.

`/pgfgo/profiles/goban/from` = ⟨*goban name*⟩ (no default)

Allows defining a goban based on a copy of another.

```
\pgfgodefinegoban[from = gobanA]{gobanB}{...}
```

It's possible to copy itself to add options, though it's not the best alternative.

```
\pgfgodefinegoban[from = gobanB]{gobanB}{...}
```

/pgfgo/profiles/goban/append (no value)

Allows expanding the options of a goban.

```
\pgfgodefinegoban[append]{gobanB}{...}
```

It is equivalent to:

```
\pgfgodefinegoban[from = gobanB]{gobanB}{...}
```

\pgfgodefineplayer[⟨*options*⟩]{⟨*player name*⟩}{⟨*player options*⟩}

Allows assigning a name to a group of stone options. Its options are equivalent to those of `\pgfgodefinegoban`.

/pgfgo/profiles/player/from = ⟨*player name*⟩ (no default)

Defines the player from a copy of another one.

/pgfgo/profiles/player/append (no value)

Extends the options of an already defined player. It is the same as `from = \meta{player name}` with its own name.

\pgfgousegoban{⟨*goban name*⟩}

Locally applies the options of the goban defined by `goban name`.

\pgfgouseneutral{⟨*player name*⟩}

Locally applies the stone options defined by `player name` to the neutral player.

\pgfgouseblack{⟨*player name*⟩}

Locally applies the stone options defined by `player name` to the black player.

\pgfgousewhite{⟨*player name*⟩}

Locally applies the stone options defined by `player name` to the white player.

\pgfgouseplayers{⟨*neutral player name*⟩}{⟨*black player name*⟩}{⟨*white player name*⟩}

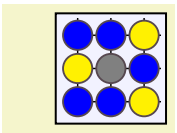
Locally applies the stone options defined by `neutral player name`, `black player name`, and `white player name` to the corresponding players.

\pgfgouseprofile{⟨*name*⟩}

This macro is equivalent to:

```
\pgfgousegoban{name}
\pgfgouseplayers{name-neutral}{name-black}{name-white}
```

Note that stones named `name-neutral`, `name-black`, and `name-white` must exist. For example, when we previously ran `\pgfgouseprofile{pgfgo}`, it worked because there is a goban called `pgfgo`, and stones named `pgfgo-neutral`, `pgfgo-black`, and `pgfgo-white`.



```
\pgfgodefinegoban{example}{background = {fill = {color = blue!05}}}
\pgfgodefineplayer[from = default-neutral]{example-neutral}
{stone = {
  fill = {color = gray},
  line = {color = magenta!20!black}}
}

\pgfgodefineplayer[from = example-neutral]{example-white}
{stone = {fill = {color = yellow}}}

\pgfgodefineplayer[from = example-neutral]{example-black}
{stone = {fill = {color = blue}}}

\pgfgouseprofile{example}

\begin{goban}[partial = {from = A1 to C3}]
 \moves{A1:3}
 \stones[b]{B1,B3,C2}
 \stones[w]{C1,C3}
 \stone{B2}
\end{goban}
```

## 3.6 Changing Options on the Fly

When you want to change a board option with `\pgfgoset`, it can be done directly with `\pgfgoset{goban = {...}}`. For players, the situation is slightly different. For this, the following *keys* are available:

`/pgfgo/players` = ⟨*players options*⟩ (no default)

Allows adjusting player options. By default, there are three: `neutral`, `black`, and `white`. Players should not be confused with stone profiles. Once a stone profile is loaded, a change can be made with these *keys*.

Player options can be configured all at once with the *key* `all`, or separately. Changes made here will not affect the profile definitions. That is, if we load the `pgfgo` profile and modify the black player, this won't affect the definition of `pgfgo-black`, but rather the player's own options first.

`/pgfgo/players/all` = ⟨*all player options*⟩ (no default)

Applies the options to the three players.

`/pgfgo/players/neutral` = ⟨*neutral player options*⟩ (no default)

Options of the neutral player (stones without a player).

`/pgfgo/players/black` = ⟨*first player options*⟩ (no default)

Options of the first player.

`/pgfgo/players/white` = ⟨*second player options*⟩ (no default)

Options of the second player.

To avoid writing the whole path, there are four commands:

`\pgfgosetplayer{`⟨*player*⟩`}{`⟨*player options*⟩`}`

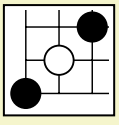
It is the same as `\pgfgoset{players = {<player> = {<player options>}}}`.

`\pgfgosetallplayers{`⟨*player options*⟩`}`

Applies the options to the three players. It is the same as `\pgfgoset{players = {all = {...}}}`.

`\pgfgocopyplayer{`⟨*target player*⟩`}{`⟨*source player*⟩`}`

Copies the current options of `source player` into `target player`.



```
\pgfgocopyplayer{neutral}{white}

\begin{goban}[partial = {from = A1 to C3}]
 \stone{B2} % Neutral, but it looks like a white stone.
 \stones[b]{A1,C3}
\end{goban}
```

`\pgfgochooseplayers{`⟨*neutral source*⟩`}{`⟨*black source*⟩`}{`⟨*white source*⟩`}`

Copies, in this order, `neutral source` into the neutral player, `black source` into the black one and `white source` into the white one (with `\pgfgocopyplayer`). Since the copies are made one after the other, it cannot be used to swap players.

Each player has access to the same key tree. For this reason, the following, although they mention `white`, apply to all players. Each is explained in the section corresponding to the object type, but they are briefly illustrated here.

`/pgfgo/players/white/stone` = ⟨*stone options*⟩ (no default)

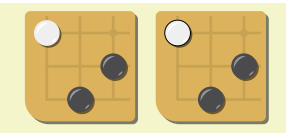
Controls the options for the second player's stones.

`/pgfgo/players/white/stone/line` = ⟨*stone line options*⟩ (no default)

Controls the options for the borders of the second player's stones.

`/pgfgo/players/white/stone/line/true` (no value)

Determines whether the stone's border appears or not.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

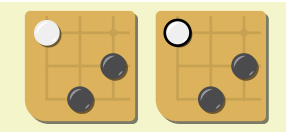
\pgfgoset{players = {white = {stone = {line = true}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

`/pgfgo/players/white/stone/line/false` (no value)

`/pgfgo/players/white/stone/line/width` = ⟨*dimension*⟩ (no default)

Determines the thickness of the stone's border.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

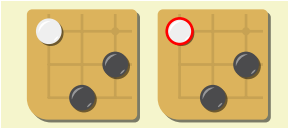
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {line = {true, width = 1pt}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

`/pgfgo/players/white/stone/line/color` = ⟨*color*⟩ (no default)

Determines the color of the stone's border.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

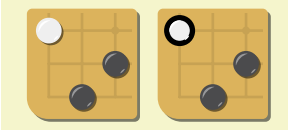
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {line = {true, width = 1pt, color = red}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/line/opacity = ⟨*float*⟩ (no default)

Determines the opacity of the stone's border.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {line = {true, width = 2pt, opacity = 0.25}}}}}

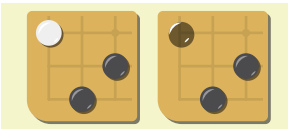
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/fill = ⟨*stone fill options*⟩ (no default)

Controls options related to the stone's fill.

/pgfgo/players/white/stone/fill/true (no value)

/pgfgo/players/white/stone/fill/false (no value)



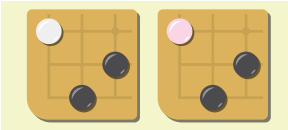
```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {fill = false}}}}
% What's visible in the figure is what remains of the stone: shadows and highlights.

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/fill/color = ⟨*color*⟩ (no default)



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

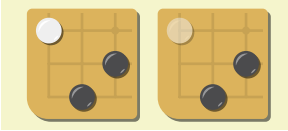
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {fill = {color = magenta!20}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

`/pgfgo/players/white/stone/fill/opacity` = ⟨*float*⟩ (no default)

Sets the opacity of the stone's fill.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

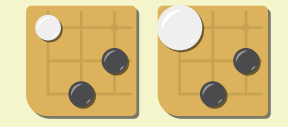
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {fill = {opacity = 0.5}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

`/pgfgo/players/white/stone/radius` = ⟨*dimension*⟩ (no default)

Sets the stone's radius.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

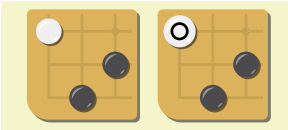
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {radius = 0.3cm}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

`/pgfgo/players/white/stone/scale` = ⟨*float*⟩ (no default)

Controls the stone's scale, which also affects markings.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {scale = 1.25}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w, mark = circle]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shadow = ⟨*stone drop shadow options*⟩ (no default)

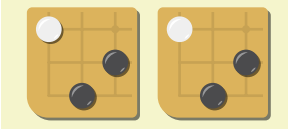
Controls options specific to shadows on stones.

/pgfgo/players/white/stone/drop shadow/true (no value)

Enables the shadow on the stone.

/pgfgo/players/white/stone/drop shadow/false (no value)

Disables the shadow on the stone.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

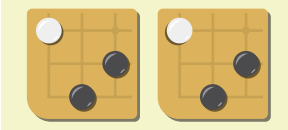
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {drop shadow = false}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shadow/angle = ⟨*float*⟩ (no default)

Sets the shadow's angle.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

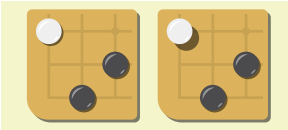
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {drop shadow = {angle = 225}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shadow/offset = ⟨*dimension*⟩ (no default)

Sets the shadow's offset from the center.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

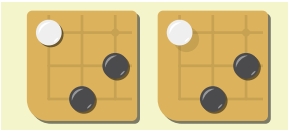
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {drop shadow = {offset = 0.125cm}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shadow/opacity = ⟨*float*⟩ (no default)

Controls the shadow's opacity.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {stone = {drop shadow = {offset = 0.125cm, opacity = 0.1}}}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

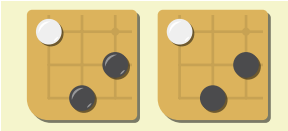
/pgfgo/players/white/stone/drop shine = ⟨*stone drop shine options*⟩ (no default)

Controls highlights on stones.

/pgfgo/players/white/stone/drop shine/true (no value)

/pgfgo/players/white/stone/drop shine/false (no value)

Hides highlights on stones.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

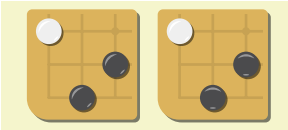
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {black = {stone = {drop shine = false}}}}
% On black, where it's more noticeable.

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shine/angle = ⟨*float*⟩ (no default)

Controls the highlight's angle.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

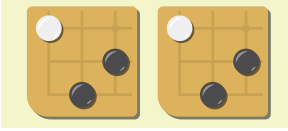
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {black = {stone = {drop shine = {angle = 0}}}}}
% On black, where it's more noticeable.

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shine/opacity = ⟨*float*⟩ (no default)

Controls the highlight's opacity.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {black = {stone = {drop shine = {opacity = 0.5}}}}}
% On black, where it's more noticeable.

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}
```

/pgfgo/players/white/stone/drop shine/color = ⟨*color*⟩ (no default, initially `gray!50!white`)

Controls the highlight's color.

/pgfgo/players/white/stone/prisoner = ⟨*stone options*⟩ (no default)

Draws the stone as a prisoner: it lowers the fill opacity to 0.5 and applies the ⟨*stone options*⟩. It is what `\prisoner` and `\prisoners` use.

/pgfgo/players/white/move = ⟨*move options*⟩ (no default)

Controls the options of the player's moves.

/pgfgo/players/white/move/label = ⟨*move label options*⟩ (no default)

Controls the labels (the numbers) of their moves.

/pgfgo/players/white/move/label/font size = ⟨*dimension*⟩ (no default)

Size of the numbers. If it is not set, it adapts to the number of digits.

/pgfgo/players/white/move/label/color = ⟨*color*⟩ (no default)

Color of the numbers.

/pgfgo/players/white/move/label/format = ⟨*format*⟩ (no default)

Format of the numbers: `arabic`, `alph`, `Alph`, `roman` or `Roman`.

/pgfgo/players/white/move/label/format/arabic (no value)

Arabic numerals (1, 2, 3).

/pgfgo/players/white/move/label/format/alph (no value)

Lowercase letters (a, b, c).

/pgfgo/players/white/move/label/format/Alph (no value)

Uppercase letters (A, B, C).

/pgfgo/players/white/move/label/format/roman (no value)

Lowercase Roman numerals (i, ii, iii).

/pgfgo/players/white/move/label/format/Roman (no value)

Uppercase Roman numerals (I, II, III).

/pgfgo/players/white/move/label/forget (no value)

The numbers of their moves are not stored in the board's memory: `resume` draws the stones again without numbers.

/pgfgo/players/white/move/label/remember (no value)

The numbers of their moves are stored in the board's memory.

/pgfgo/players/white/mark = ⟨*mark options*⟩ (no default)

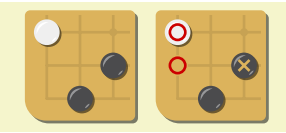
Controls the options for markings associated with the corresponding player.

/pgfgo/players/white/mark/line = ⟨*mark line options*⟩ (no default)

Controls the line of their marks.

/pgfgo/players/white/mark/line/color = ⟨*color*⟩ (no default)

Controls the line color of markings.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

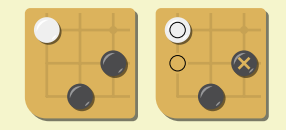
\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {mark = {line = {color = red!80!black}}}}}

\begin{goban}
 \stones[b] {B1}
 \stone [b, mark] {C2}
 \stone [w, mark = circle] {A3}
 \mark  [w, circle] {A2}
\end{goban}
```

/pgfgo/players/white/mark/line/width = ⟨*dimension*⟩ (no default)

Controls the line thickness of markings.



```
\pgfgouseprofile{pgfgosmooth}
\pgfgoset{goban = {partial = {from = A1 to C3}}}

\begin{goban}
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\pgfgoset{players = {white = {mark = {line = {width = 0.3pt}}}}}

\begin{goban}
 \stones[b] {B1}
 \stone [b, mark] {C2}
 \stone [w, mark = circle] {A3}
 \mark  [w, circle] {A2}
\end{goban}
```

`/pgfgo/players/white/mark/fill` = ⟨*mark fill options*⟩ (no default)

Controls the fill of their filled marks (`filled square`, `filled circle` and `hatched lines`).

`/pgfgo/players/white/mark/fill/true` (no value)

Filled marks use `fill/color`.

`/pgfgo/players/white/mark/fill/false` (no value)

Filled marks are filled in gray. It is the initial value.

`/pgfgo/players/white/mark/fill/color` = ⟨*color*⟩ (no default)

Fill color of the marks. It is only used with `fill = true`.

`/pgfgo/players/white/mark/label` = ⟨*mark label options*⟩ (no default)

Controls label options for markings that have them (`sequence` and `custom`).

`/pgfgo/players/white/mark/label/color` = ⟨*color*⟩ (no default)

Color of the labels of their marks.

`/pgfgo/players/white/mark/label/font` = ⟨*font commands*⟩ (no default)

Font of the labels of their marks.

`/pgfgo/players/white/mark/label/format` = ⟨*format*⟩ (no default)

Format of the numbers of `sequence`: `arabic`, `alph`, `Alph`, `roman` or `Roman`.

`/pgfgo/players/white/mark/label/format/arabic` (no value)

Arabic numerals (1, 2, 3).

`/pgfgo/players/white/mark/label/format/alph` (no value)

Lowercase letters (a, b, c).

`/pgfgo/players/white/mark/label/format/Alph` (no value)

Uppercase letters (A, B, C).

`/pgfgo/players/white/mark/label/format/roman` (no value)

Lowercase Roman numerals (i, ii, iii).

`/pgfgo/players/white/mark/label/format/Roman` (no value)

Uppercase Roman numerals (I, II, III).

`/pgfgo/players/white/mark/sequence` =⟨*mark sequence options*⟩ (default `from = 1`)

Numbers their marks in order.

`/pgfgo/players/white/mark/sequence/true` (no value)

Turns the sequence on.

`/pgfgo/players/white/mark/sequence/use` (no value)

The same as `sequence/true`.

`/pgfgo/players/white/mark/sequence/false` (no value)

Turns the sequence off.

`/pgfgo/players/white/mark/sequence/from` = ⟨*number*⟩ (no default)

Turns the sequence on and starts it at ⟨*number*⟩.

`/pgfgo/players/white/mark/sequence/format` = ⟨*format*⟩ (no default)

Turns the sequence on and chooses the number format (the same as `label/format`).

`/pgfgo/players/white/mark/custom` = ⟨*token list*⟩ (no default)

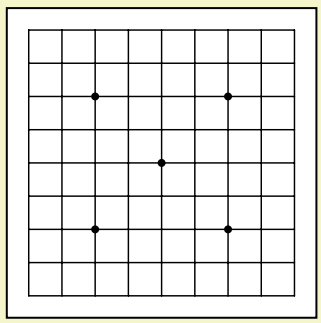
Custom mark with text or symbols, like `/pgfgo/mark/custom`.

# 4 The Board

When we refer to the ”board,” we mean an environment or *scope* in which both the goban (the game grid) and its contents—stones, coordinate labels, and other relevant information—are rendered. This environment is designed to be flexible and customizable, allowing users to adapt its appearance and functionality according to the needs of the diagram, whether to represent full games, specific sections, or even add visual annotations.

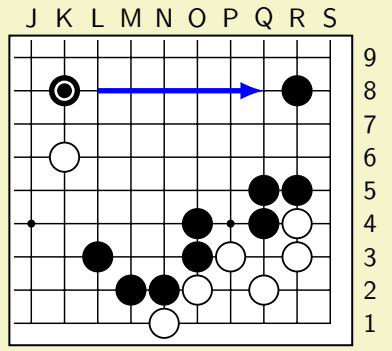
`\begin{goban}[`⟨*options*⟩`]`
  ⟨*environment contents*⟩
`\end{goban}`

Initializes the goban environment. By default, it draws a square 9×9 intersection board with a visible border, white background, and no labels on the sides.



```
\begin{goban}
\end{goban}
```

Since the board is the only macro that utilizes the TikZ layer, each intersection can be accessed using its alphanumeric coordinate (as in PGF) or numeric values in the Ti*k*Z coordinate system. This allows combining Go's intuitive syntax with TikZ's graphical capabilities, which is especially useful for adding custom elements, such as arrows or marks, directly onto the board.

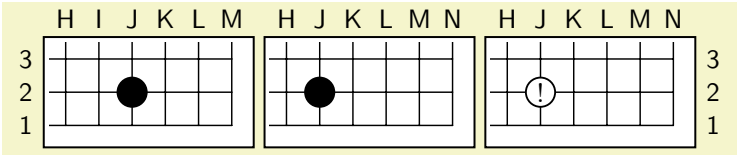



```
\begin{goban}[19, partial = {from = J1 to S9},
 label = {at = {north, east}}]
 \stones {K8,R8,Q5,R5,O4,Q4,L3,O3,M2,N2}
 \mark[black, circle] {K8}
 \stones[white] {K6,R4,R3,P3,N1,O2,Q2}

 \tikzset{flecha/.style = {-{latex[]}, line width = 2pt, blue}}
 \draw[flecha] (12,8)--(17,8);
\end{goban}
```

`/pgfgo/goban/use i=`⟨*boolean*⟩ (default `true`, initially `true`)

Enables or disables the use of the letter ”I” in the board's coordinates and labels. When disabled (”false”), any reference to ”I” is interpreted as ”J”.



```
\pgfgoset{goban = {13, label = {at = north}}}

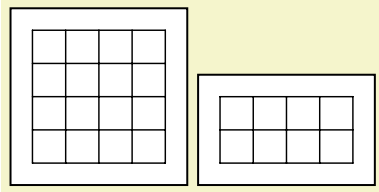
\begin{goban}[label = {at = west}, partial = {from = H1 to M3}]
 \stone[b]{J2}
\end{goban}

\begin{goban}[use i = false, partial = {from = H1 to N3}]
 \stone[b]{J2}
\end{goban}

\begin{goban}[use i = false, label = {at = east}, partial = {from = H1 to N3}]
 \stone[white, mark = {custom = !}] {I2} % If it's I, it shifts to J.
\end{goban}
```

/pgfgo/goban/size=⟨*nlines or nlines* x *nlines*⟩ (no default, initially 9)

Defines the number of vertical and horizontal lines on the board. The word "size" can be omitted, and the value specified directly. If a single integer (e.g., 9) is provided, the board will be square (9×9). If an expression like 5x3 is used, a rectangular board with 5 columns and 3 rows will be created.



```
\begin{goban}[5]
\end{goban}

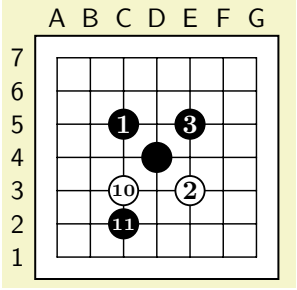
\begin{goban}[5x3]
\end{goban}
```

/pgfgo/goban/sgf view=⟨*file*⟩ (no default)

Like sgf size and, if the file has a view (VW), it only shows that part of the board, like partial.

/pgfgo/goban/sgf size=⟨*file*⟩ (no default)

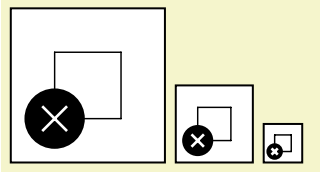
Takes the size of the board from an SGF game (its SZ property). Only with the Lua kernel (p. 12).




```
\begin{goban}[sgf size = variations.sgf, label = {at = {north, west}}]
 \sgf{variations.sgf}
\end{goban}
```

/pgfgo/goban/scale=⟨*float*⟩ (no default, initially 1)

Controls the global scale of the board, affecting both its overall dimensions and the thickness of certain elements (e.g., the "line width" of the lines).



```
\begin{goban}[2, scale = 2]
 \stone[b, mark]{A1}
\end{goban}

\begin{goban}[2, scale = 1]
 \stone[b, mark]{A1}
\end{goban}

\begin{goban}[2, scale = 0.5]
 \stone[b, mark]{A1}
\end{goban}
```

## 4.1 Grid

The grid refers to the central lines that form the board's intersections.

`/pgfgo/goban/grid`=⟨*options*⟩ (no default)

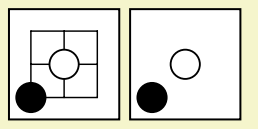
Controls the behavior and appearance of the board's central lines. This set of options allows modifying their visibility, color, thickness, and spacing, offering a high degree of customization.

`/pgfgo/goban/grid/true` (style, no value)

Turns the central lines back on after a `grid = false` (for example, a global one set with `\pgfgoset`).

`/pgfgo/goban/grid/false` (style, no value)

Disables the central lines, leaving only the border and stones visible.



```
\begin{goban}[3]
 \stone[b]{A1}
 \stone[w]{B2}
\end{goban}

\begin{goban}[3, grid = false]
 \stone[b]{A1}
 \stone[w]{B2}
\end{goban}
```

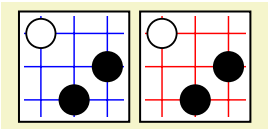
The first board shows the default grid.

`/pgfgo/goban/grid/line`=⟨*options*⟩ (no default)

Allows controlling the grid lines.

`/pgfgo/goban/grid/line/color` =⟨*color*⟩ (no default, initially `black`)

Allows changing the color of the grid lines.



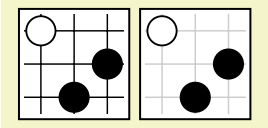
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[grid = {line = {color = blue}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[grid = {line = {color = red}}, resume]
\end{goban}
```

`/pgfgo/goban/grid/line/opacity` =⟨*float*⟩ (no default, initially `1`)

Sets the opacity of the grid lines, from 0 (transparent) to 1 (opaque).



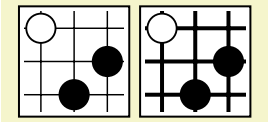
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[grid = {line = {opacity = 0.2}}, resume]
\end{goban}
```

`/pgfgo/goban/grid/line/width` =⟨*dimension*⟩ (no default, initially `0.55pt`)

Defines the thickness of the grid lines.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[grid = {line = {width = 1.5pt}}, resume]
\end{goban}
```

/pgfgo/goban/grid/x sep =⟨*dimension*⟩ (no default, initially `1.25em`)

Sets the spacing between vertical lines, adjusting the width of the board's columns.

/pgfgo/goban/grid/y sep =⟨*dimension*⟩ (no default, initially `1.25em`)

Sets the spacing between horizontal lines, adjusting the height of the rows.

/pgfgo/goban/grid/sep =⟨*dimension*⟩ (no default, initially `1.25em`)

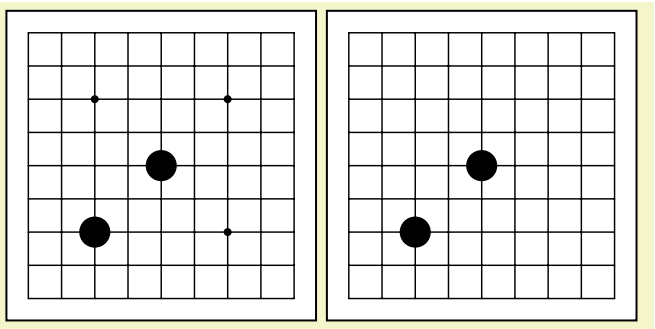
Simultaneously defines the vertical and horizontal spacing of the lines, ensuring uniform spacing in both directions. This is a convenient option for maintaining balanced proportions.

## 4.2 Star Points

The star points (*hoshi*) are drawn on the grid, in its color. By default they are the usual ones: on each side, the fourth line from the edge (the third on boards with fewer than 13 lines); the center if the size is odd; and, from 15 lines on, the middle of the sides. A 19 board has nine, and a 13 or 9 board, five. Boards with fewer than 7 lines have none. Stones cover them.

/pgfgo/goban/hoshi=⟨*options*⟩ (default `true`)

Options of the star points.



```
\begin{goban}[9]
 \stones{C3, E5}
\end{goban}

\begin{goban}[9, hoshi = false]
 \stones{C3, E5}
\end{goban}
```

/pgfgo/goban/hoshi/true (style, no value)

Draws the star points. It is the initial value.

/pgfgo/goban/hoshi/false (style, no value)

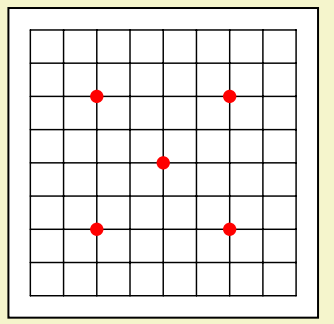
Does not draw the star points.

/pgfgo/goban/hoshi/radius=⟨*dimension*⟩ (no default, initially `1.5pt`)

Radius of the points.

/pgfgo/goban/hoshi/color=⟨*color*⟩ (no default)

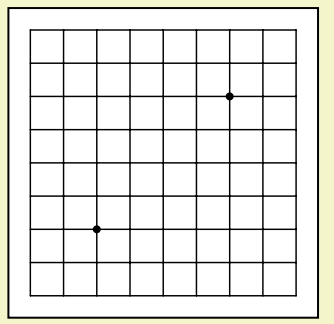
Color of the points. Without this *key*, they take the color of the grid.



```
\begin{goban}[9, hoshi = {radius = 2.5pt, color = red}]
\end{goban}
```

/pgfgo/goban/hoshi/at=⟨*coordinates*⟩ (no default)

Chooses the points, with the same syntax as `\stones`. Empty, the usual ones are used.



```
\begin{goban}[9, hoshi = {at = {C3, G7}}]
\end{goban}
```

## 4.3 Border

The board's border is the outline that frames the grid and stones, providing a clear visual boundary. By default, it is active, has a white background, and a black outline.

/pgfgo/goban/background=⟨*options*⟩ (no default)

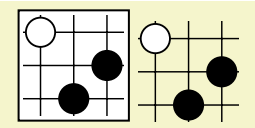
Controls the properties of the board's border, such as its visibility, color, thickness, and shape, allowing detailed customization.

/pgfgo/goban/background/true (style, no value)

Turns the board's background and border back on after a `background = false`.

/pgfgo/goban/background/false (style, no value)

Completely disables the board's border, removing both the outline and fill. This is useful for minimalist diagrams or when the document's background already provides sufficient contrast.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

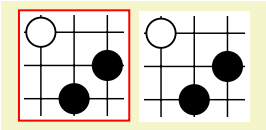
\begin{goban}[background = false, resume]
\end{goban}
```

/pgfgo/goban/background/fill=⟨*options*⟩ (no default)

Controls options related to the board's background fill.

/pgfgo/goban/background/fill=⟨*boolean*⟩ (no default, initially `true`)

Determines whether the board's background fill is active. If disabled, the area inside the border will be transparent, showing only the outline (if enabled).



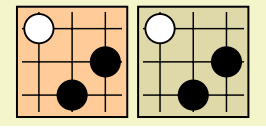
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {line = {color = red}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {line = false}, resume]
\end{goban}
```

/pgfgo/goban/background/fill/color =⟨*color*⟩ (no default, initially `white`)

Defines the fill color of the board.



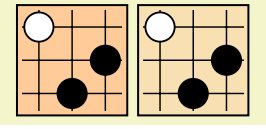
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {fill = {color = orange!40}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {fill = {color = olive!30}}, resume]
\end{goban}
```

/pgfgo/goban/background/fill/opacity =⟨*float*⟩ (no default, initially `1`)

Controls the opacity of the fill, from 0 (transparent) to 1 (opaque).



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {fill = {color = orange!40}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

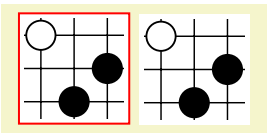
\begin{goban}[background = {fill = {color = orange!40}, fill = {opacity = 0.5}}, resume]
\end{goban}
```

/pgfgo/goban/background/line=⟨*options*⟩ (no default)

Controls the options for the background's outline.

/pgfgo/goban/background/line=⟨*boolean*⟩ (no default, initially `true`)

Enables or disables the border line.



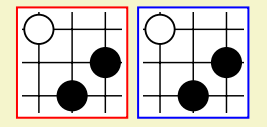
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {line = {color = red}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {line = false}, resume]
\end{goban}
```

/pgfgo/goban/background/line/color =⟨*color*⟩ (no default, initially `black`)

Defines the color of the border line, allowing it to stand out from the background or grid lines.



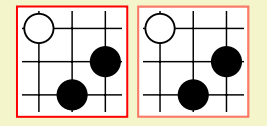
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {line = {color = red}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {line = {color = blue}}, resume]
\end{goban}
```

/pgfgo/goban/background/line/opacity =⟨*float*⟩ (no default, initially `1`)

Adjusts the opacity of the border line.



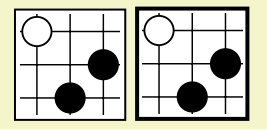
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {line = {color = red}}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {line = {color = red}, line = {opacity = 0.5}}, resume]
\end{goban}
```

/pgfgo/goban/background/line/width =⟨*dimension*⟩ (no default, initially `0.75pt`)

Controls the thickness of the border line.



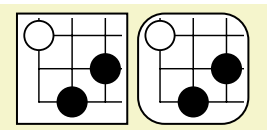
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {line = {width = 1.5pt}}, resume]
\end{goban}
```

/pgfgo/goban/background/rounded corners =⟨*dimension*⟩ (no default, initially `0pt`)

Adjusts the rounding of the border corners, from 0pt (straight corners) to larger values for a smoother, more stylized effect.



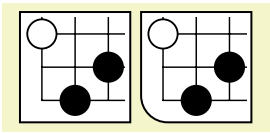
```
\pgfgoset{goban = {5, partial = {from = A1 to C3}}}

\begin{goban}[new remember]
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\begin{goban}[background = {rounded corners = 10pt}, resume]
\end{goban}
```

/pgfgo/goban/background/rounded corners/outer =⟨*dimension*⟩ (no default, initially `0pt`)

Adjusts only the rounding of the outer corners of the border.



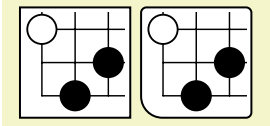
```
\pgfgoset{goban = {5, partial = {from = A1 to C3}}}

\begin{goban}[background = {rounded corners = 0pt}, new remember]
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\begin{goban}[background = {rounded corners = {outer = 10pt}}, resume]
\end{goban}
```

/pgfgo/goban/background/rounded corners/inner =⟨*dimension*⟩ (no default, initially `0pt`)

Adjusts only the rounding of the inner corners of the board (only visible on partial boards).



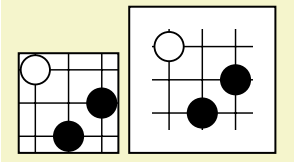
```
\pgfgoset{goban = {5, partial = {from = A1 to C3}}}

\begin{goban}[background = {rounded corners = 0pt}, new remember]
 \stones[b]{B1,C2}
 \stone[w]{A3}
\end{goban}

\begin{goban}[background = {rounded corners = {outer = 8pt, inner = 2pt}}, resume]
\end{goban}
```

/pgfgo/goban/background/sep =⟨*dimension*⟩ (no default, initially `0.825em`)

Defines the separation between the grid and the border, based by default on the stone radius multiplied by 1.5. This adjustment ensures that stones near the border do not overlap with it.



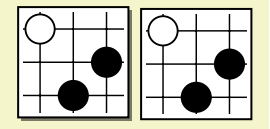
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {sep = 0.625em}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {sep = 1.5em}, resume]
\end{goban}
```

/pgfgo/goban/background/drop shadow=⟨*options*⟩ (default `true`)

Draws a shadow under the board. Without a value, it turns it on.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {drop shadow}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

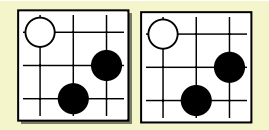
\begin{goban}[background = {drop shadow = {color = blue, opacity = 0.3}}, resume]
\end{goban}
```

/pgfgo/goban/background/drop shadow=⟨*boolean*⟩ (no default, initially `false`)

Turns the shadow on (`true`) or off (`false`).

/pgfgo/goban/background/drop shadow/angle =⟨*float*⟩ (no default, initially `-45`)

Direction of the shadow, in degrees.



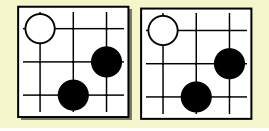
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {drop shadow}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {drop shadow = {angle = 45}}, resume]
\end{goban}
```

/pgfgo/goban/background/drop shadow/offset =⟨*dimension*⟩ (no default, initially `1.5pt`)

Distance between the board and its shadow.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[background = {drop shadow}, new remember]
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}[background = {drop shadow = {offset = 4pt}}, resume]
\end{goban}
```

/pgfgo/goban/background/drop shadow/opacity =⟨*float*⟩ (no default, initially `0.5`)

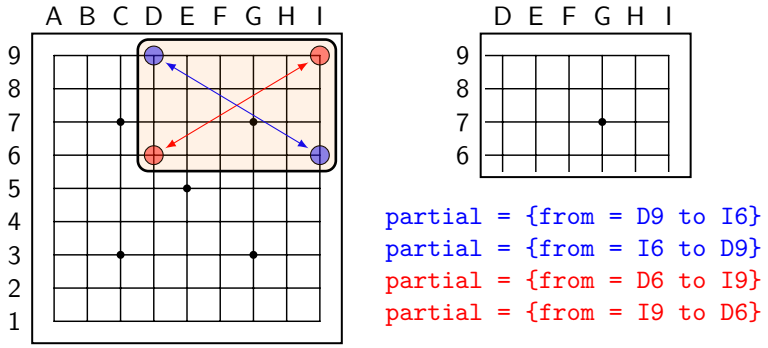
Opacity of the shadow.

/pgfgo/goban/background/drop shadow/color =⟨*color*⟩ (no default, initially `black`)

Color of the shadow.

## 4.4 Partial Board

In many cases, it's not necessary to represent a full board, but rather just a rectangular section—whether due to space constraints, to focus attention on a specific area, or to simplify diagram creation. The `goban` environment includes the *key* `partial`, which allows defining this region using the sub-*key* `from`. This specifies the opposite vertices of a rectangle, regardless of the order in which they are provided, making it easy to use. For example, `from = A1 to D3` is equivalent to `from = D3 to A1`.

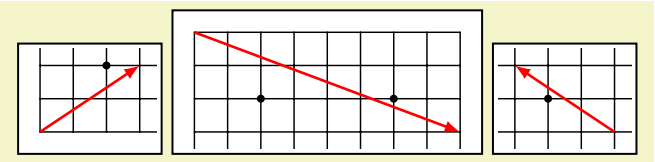



/pgfgo/goban/partial=⟨*options*⟩ (no default)

Controls options for representing partial boards, allowing the delimitation of a specific region of the goban.

/pgfgo/goban/partial/from=⟨*intersection* `to` *intersection*⟩ (no default)

Specifies the opposite vertices of the rectangle defining the partial board. These intersections can be any pair of points forming a diagonal of the desired area.



```
\tikzset{flecha/.style = {-{latex[]}, line width = 1pt, red}}

\begin{goban}[partial = {from = A1 to D3}]
 \draw[flecha] (1,1)--(4,3); % From A1 to D3
\end{goban}

\begin{goban}[partial = {from = A9 to I6}]
 \draw[flecha] (1,9)--(9,6); % From A9 to I6
\end{goban}

\begin{goban}[partial = {from = E2 to B4}]
 \draw[flecha] (5,2)--(2,4); % From E2 to B4
\end{goban}
```

/pgfgo/goban/partial/false (no value)

Shows the whole board again (for example, after a global `partial` set with `\pgfgoset`).

## 4.5 Board Labels

Labels are the alphanumeric indicators (letters and numbers) that appear on the sides of the board to identify intersections.

/pgfgo/goban/label=⟨*options*⟩ (no default)

Controls the properties of the labels, such as their position and format.

/pgfgo/goban/label/false (no value)

Disables all labels.

/pgfgo/goban/label/at =⟨*options*⟩ (no default)

Defines on which sides of the board the labels appear. Options include combinations of cardinal positions or predefined values like ”all” or ”none”.

/pgfgo/goban/label/at/all (style, no value)

Displays labels on all sides of the board (north, south, east, and west).

/pgfgo/goban/label/at/none (style, no value)

Equivalent to `label=false`, removing all labels.

/pgfgo/goban/label/at/east (style, no value)

Places labels only on the right side (vertical numbers).

/pgfgo/goban/label/at/right (style, no value)

Equivalent to `east`.

/pgfgo/goban/label/at/north (style, no value)

Places labels only at the top (horizontal letters).

/pgfgo/goban/label/at/above (style, no value)

Equivalent to `north`.

/pgfgo/goban/label/at/south (style, no value)

Places labels only at the bottom (horizontal letters).

/pgfgo/goban/label/at/below (style, no value)

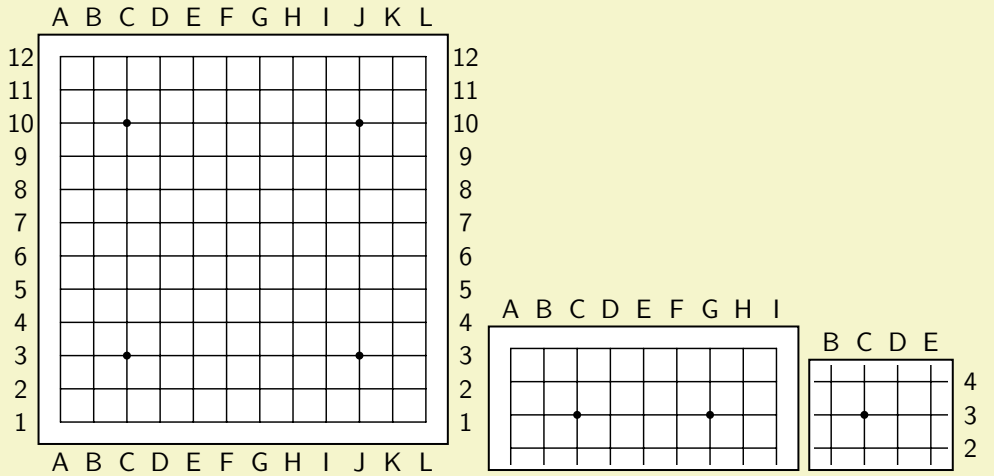
Equivalent to `south`.

/pgfgo/goban/label/at/west (style, no value)

Places labels only on the left side (vertical numbers).

/pgfgo/goban/label/at/left (style, no value)

Equivalent to `west`.



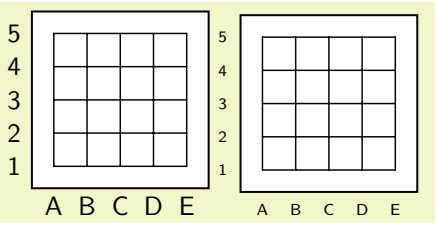

```
\begin{goban}[12, label = {at = {all}}]
\end{goban}

\begin{goban}[partial = {from = A6 to I9}, label = {at = north}]
\end{goban}

\begin{goban}[partial = {from = B2 to E4}, label = {at = {north, east}}]
\end{goban}
```

/pgfgo/goban/label/font size = ⟨*dimension*⟩ (no default, initially `the current font size`)

Adjusts the font size of the labels.

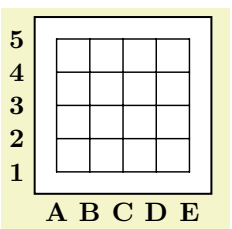



```
\begin{goban}[5, label = {at = {south, west}}]
\end{goban}

\begin{goban}[5, label = {at = {south, west}, font size = 6pt}]
\end{goban}
```

/pgfgo/goban/label/font = ⟨*font commands*⟩ (no default, initially `\sffamily`)

Sets the font of the labels.




```
\begin{goban}[5, label = {at = {south, west}, font = \bfseries}]
\end{goban}
```

/pgfgo/goban/label/sep = ⟨*dimension*⟩ (no default, initially `0.35cm`)

Controls the distance between the labels and the board's border.

/pgfgo/goban/label/text color = ⟨*color*⟩ (no default, initially `black`)

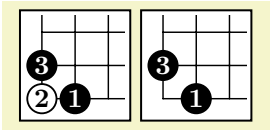
Defines the text color of the labels.

## 4.6 Captures

With the Lua kernel (p. 12), PGFGO can keep track of the stones on the board and remove the ones left without liberties, as in a game.

/pgfgo/goban/captures=⟨*remove, fade or false*⟩ (default `remove`, initially `false`)

Turns captures on. When a move leaves an opposing group without liberties, that group is removed from the board. If it leaves its own group without liberties (a suicide), that group is removed. Stones placed with `\stone` and `\stones` also count; neutral stones and prisoners do not. A move outside a partial board also captures. A move on an occupied point produces a warning, because it is not possible in a game. With `kernel = expl3`, this *key* raises an error.



```
\pgfgoset{goban = {5, partial = {from = A1 to C3}}}

\begin{goban}
 \moves*{B1,A1,A2}
\end{goban}

\begin{goban}[captures]
 \moves*{B1,A1,A2} % 3 captures 2.
\end{goban}
```

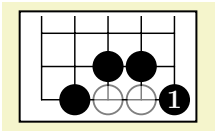
Since PGF cannot erase what it has already drawn, with `captures` the stones are drawn when the `goban` ends. For this reason, if the style is changed with `\pgfgoset` inside a `goban` with captures, the change does not reach the stones; the options of each command do.

/pgfgo/goban/captures/remove (style, no value)

Captured stones disappear. It is the same as `captures` or `captures = true`.

/pgfgo/goban/captures/fade (style, no value)

Captured stones stay on the board, drawn as prisoners (with the style of `\prisoner` and `/pgfgo/prisoner`).



```
\pgfgoset{goban = {5, partial = {from = A1 to E3}}}

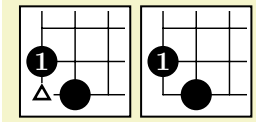
\begin{goban}[captures = fade]
 \stones[w]{C1,D1}
 \stones[b]{B1,C2,D2}
 \moves*{E1}
\end{goban}
```

/pgfgo/goban/captures/false (style, no value)

Turns captures off. It is the initial value.

/pgfgo/goban/captured marks=⟨*keep or remove*⟩ (no default, initially `keep`)

Decides what happens to the marks on a stone when it is captured. It only works with `captures`. Marks on empty points do not change.



```
\pgfgoset{goban = {5, partial = {from = A1 to C3}, captures}}

\begin{goban}
 \stones[w]{A1}\stones{B1}
 \marks[triangle]{A1}
 \moves*{A2}
\end{goban}

\begin{goban}[captured marks = remove]
 \stones[w]{A1}\stones{B1}
 \marks[triangle]{A1}
 \moves*{A2} % the triangle goes with the white stone.
\end{goban}
```

/pgfgo/goban/captured marks/keep (style, no value)

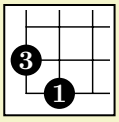
The marks stay on the empty point. It is the initial value.

/pgfgo/goban/captured marks/remove (style, no value)

The marks go with the stone, also with `captures = fade`. Sequences keep their numbers.

`\pgfgoprisoners{`⟨*player*⟩`}`

Writes how many prisoners the player (`black` or `b`, `white` or `w`) has in the last `goban` with `captures`: the opponent's stones that their moves captured.


Black: 1. White: 0.

```
\pgfgoset{goban = {5, partial = {from = A1 to C3}}}

\begin{goban}[captures]
 \moves*{B1,A1,A2}
\end{goban}

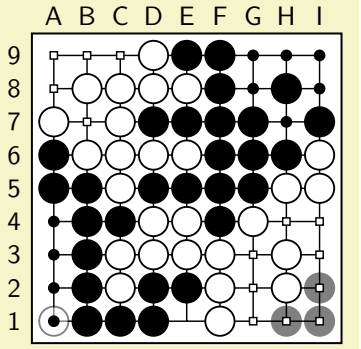
Black: \pgfgoprisoners{black}. White: \pgfgoprisoners{white}.
```

## 4.7 Territory

At the end of the game, the territory can be marked by hand with `\marks` (p. 61) or with `\territory`, which also counts it.

`\territory[`⟨*options*⟩`]{`⟨*polygon*⟩`}`

Marks as territory the points of the polygon with those vertices (in order), including its border. It does not mark the points that have a living stone: prisoners count as territory. Vertices are written as in `\stones`; a single one marks a point, and two, a segment. It works with both kernels.

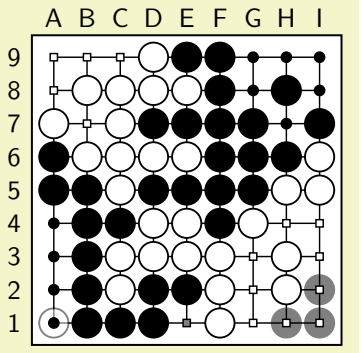


```
\begin{goban}[label = {at = {above, left}}]
\stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
\stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}
\prisoners[b]{H1-|I2}
\prisoner[w] {A1}

  \territory[b]{A1,A5,B5,B1}
  \territory[b]{F9,F7,G7,G6,I6,I9}
  \territory[w]{F1,F4,G4,G5,I5,I1}
  \territory[w]{A9,A6,C6,C8,D8,D9}
\end{goban}
```

`\territory[`⟨*options*⟩`]{`⟨*auto*⟩`}`

With `auto` instead of a polygon, it finds the territory by itself: each region of points without living stones belongs to the color around it. If both colors touch it (dame, seki), or none, it is neutral. Prisoners count as dead stones. Without a player, it marks the territory of both; with `b` or `w`, only that player's. Only with the Lua kernel (p. 12).




```
\begin{goban}[label = {at = {above, left}}]
\stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
\stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}
\prisoners[b]{H1-|I2}
\prisoner[w] {A1}

  \territory[dame]{auto}
\end{goban}
```

/pgfgo/territory/black (no value)

The territory belongs to Black. It can also be written `b`.

/pgfgo/territory/white (no value)

The territory belongs to White. It can also be written `w`.

/pgfgo/territory/neutral (no value)

Neutral points. It is what a polygon without a player marks. It can also be written `n`.

/pgfgo/territory/b (no value)

Short for `black`.

/pgfgo/territory/w (no value)

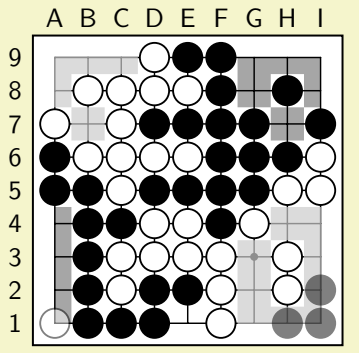
Short for `white`.

/pgfgo/territory/n (no value)

Short for `neutral`.

/pgfgo/territory/draw=⟨*marks or fill*⟩ (no default, initially `marks`)

With `marks`, every point gets the territory mark of the player. With `fill`, the region is filled below the stones.




```
\begin{goban}[label = {at = {above, left}}]
\stones[b] {A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, I7}
\stones[w] {E8-|C5, B:E6, A7, B8, D9, I6|- H5, H2:3, F1|-C3, C2, D:E4, G4}
\prisoners[b]{H1-|I2}
\prisoner[w] {A1}

 \territory[draw = fill]{auto}
\end{goban}
```

`/pgfgo/territory/dame`=⟨*boolean*⟩ (default `true`, initially `false`)

With `auto`, it also marks the neutral points.

`/pgfgo/territory/remember` (no value)

This territory is saved in the memory of the board, even if the `goban` uses `forget`.

`/pgfgo/territory/forget` (no value)

This territory is not saved in the memory of the board, even if the `goban` uses `remember`.

The other options of `\territory` change the style of the player only for that command, like `mark` and `fill` in `\territory[w, mark = {cross}]{auto}`. The style of each player is set with `/pgfgo/players`:

`/pgfgo/players/black/territory`=⟨*territory options*⟩ (no default)

Style of the territory of each player (`black`, `white` or `neutral`).

`/pgfgo/players/black/territory/mark`=⟨*mark options*⟩ (no default, initially `t-circle`)

Mark of every point, with `draw = marks`. It is `t-circle` for Black and `t-square` for White and the neutral points.

`/pgfgo/players/black/territory/fill`=⟨*fill options*⟩ (no default)

Fill, with `draw = fill`.

`/pgfgo/players/black/territory/fill/color`=⟨*color*⟩ (no default, initially `black`)

Color of the fill: `black` for Black, `white!80!black` for White and `gray` for the neutral points.

`/pgfgo/players/black/territory/fill/opacity`=⟨*number*⟩ (no default, initially `0.35`)

Opacity of the fill: `0.35` for Black and the neutral points, `0.7` for White.

`\pgfgoscore[`⟨*options*⟩`]{`⟨*player*⟩`}`

Writes the score of the player (`black` or `b`, `white` or `w`) in the last `goban`, with Japanese rules: the points of their territory, plus the opponent's dead stones in it, plus the stones they captured (with `captures`), plus the `komi` for White. A point marked twice counts once.

```
Black: \pgfgoscore{black}. White: \pgfgoscore[komi = 6.5]{white}.
```

With the board above, it writes "Black: 11. White: 23.5".

/pgfgo/score/komi=⟨*number*⟩ (no default, initially 0)

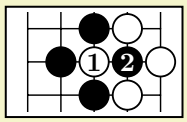
Komi added to the score of White. It can be set for the whole document with `\pgfgoset{score = {komi = 6.5}}`.

## 4.8 Moves on Points Already Played

In numbered diagrams, when a move is played on a point where there already was a move (for example, when a ko is retaken), books do not draw it: the point still shows the first stone, and below the diagram they write, for example, ”3 at 1”.

/pgfgo/goban/notes=⟨*boolean*⟩ (default `true`, initially `false`)

A numbered move on a point where there already was a numbered move of the same `goban` is not drawn: the point still shows the first one, even if it was captured, and ”N at M” is noted. Only numbered moves (`\moves*`) that are visible on the board count. It works with both kernels.



3 at 1

```
\pgfgoset{goban = {5, partial = {from = A2 to E4}}}

\begin{goban}[notes]
 \stones{B3,C2,C4}
 \stones[w]{D2,D4,E3}
 \moves*[w]{C3}
 \moves*{D3}
 \moves*[from = 3, w]{C3}
\end{goban}

\pgfgomovenotes
```

`\pgfgomovenotes[`⟨*options*⟩`]`

Writes the notes of the last `goban`, separated by commas. If there are no notes, it writes nothing. The options are those of `/pgfgo/notes`.

/pgfgo/notes = ⟨*options*⟩ (no default)

Options of `\pgfgomovenotes` for the whole document, for example `\pgfgoset{notes = {at = en}}`.

/pgfgo/notes/at = ⟨*text*⟩ (no default, initially `at`)

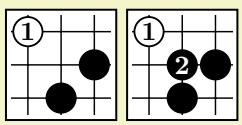
Text between the two numbers.

/pgfgo/notes/separator = ⟨*text*⟩ (no default, initially `a comma and a space`)

Text between the notes.

## 4.9 Remembering a Board

The ”remember” system stores what is drawn in a `goban` (stones, moves and marks, with their options) to draw it again in another one with `resume`. Moves keep their number and color, and the numbering continues from there. Only what is drawn is stored: the board options (size, `partial`, labels) are given in each `goban`.




```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \stones[b]{C2,D3}
 \moves*[w]{B4}
\end{goban}

\begin{goban}[resume]
 \moves*{C3} % Continues with 2 (Black).
\end{goban}
```

/pgfgo/goban/new remember (style, no value)

Empties the memory and stores what is drawn in this `goban`.

/pgfgo/goban/remember (style, no value)

Adds what is drawn in this `goban` to the memory, without erasing what was there.

/pgfgo/goban/add to remember (style, no value)

The same as `remember`.

/pgfgo/goban/forget (style, no value)

Empties the memory and stores nothing from this `goban`, except what is drawn with the option `remember` (for example, `\stone[remember]{C3}`).

/pgfgo/goban/resume (style, no value)

Draws what is in the memory at the beginning of the `goban`.

# 5 Stones, Moves, and Prisoners

Stones are the fundamental elements. In PGFGO, commands and options are provided to draw them flexibly, allowing customization of their appearance and behavior according to the needs of the diagram.

## 5.1 Stones

`\stone[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

This command draws an individual stone at the specified intersection. The mandatory argument is the coordinate (e.g., "C2"), while optional options allow adjusting its style, such as color, size, or markings. If no player is given (`b`, `w` or `n`), the stone is *neutral*.

`\stones[`⟨*options*⟩`]{`⟨*intersection , intersection, ...*⟩`}`

Similar to `\stone`, but allows placing multiple stones using the corresponding syntax (see Coordinate Syntax). If no player is given, the stones are *black*.

Stones, marks and labels that fall outside the visible region of a partial board are not drawn. A coordinate that does not exist on the board (for example, `E9` on a 5×5 board) is ignored and produces a warning.

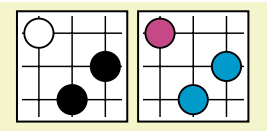
It is possible to change an option for all players using `\pgfgoset{stone = {...}}`. This is equivalent to `\pgfgoset{players = {all = {stone = {...}}}}`.

/pgfgo/stone/fill = ⟨*options*⟩ (no default)

Controls the fill options of a stone.

/pgfgo/stone/fill/color = ⟨*color*⟩ (no default)

Defines the fill color of a stone.



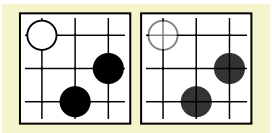
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[fill = {color = cyan!80!black}]{C2,D3}
 \stone[fill = {color = magenta!80!black}]{B4}
\end{goban}
```

/pgfgo/stone/fill/opacity = ⟨*float*⟩ (no default, initially 1)

Controls the opacity of the stone's fill, ranging from 0 (transparent) to 1 (opaque).



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

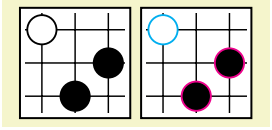
\begin{goban}
 \stones[b, fill = {opacity = 0.8}]{C2,D3}
 \stone[w, fill = {opacity = 0.5}]{B4}
\end{goban}
```

/pgfgo/stone/line = ⟨*option*⟩ (no default, initially `true`)

Controls the border options of a stone.

/pgfgo/stone/line/color = ⟨*color*⟩ (no default, initially `black`)

Sets the border color of the stone, which is black by default for all stones.



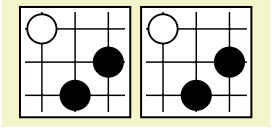
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b, line = {color = magenta}]{C2,D3}
 \stone[w, line = {color = cyan}]{B4}
\end{goban}
```

/pgfgo/stone/line/opacity = ⟨*float*⟩ (no default, initially `1`)

Defines the opacity of the stone's border, allowing it to be more subtle or transparent.



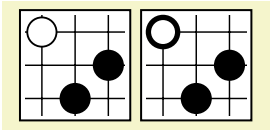
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b, line = {opacity = 0.5}]{C2,D3}
 \stone[w, line = {opacity = 0.8}]{B4}
\end{goban}
```

/pgfgo/stone/line/width = ⟨*dimension*⟩ (no default, initially `0.7pt`)

Controls the thickness of the stone's border.



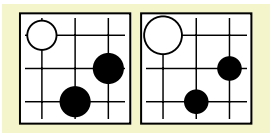
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w, line = {width = 2pt}]{B4}
\end{goban}
```

/pgfgo/stone/radius = ⟨*dimension*⟩ (no default, initially `5.5pt`)

Sets the size of the stone by its radius.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

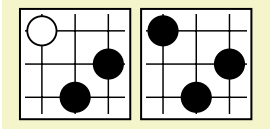
\begin{goban}
 \stones[b, radius = 0.15cm]{C2,D3}
 \stone[w, radius = 0.25cm]{B4}
\end{goban}
```

/pgfgo/stone/scale = ⟨*float*⟩ (no default, initially 1)

Defines a scaling factor that affects all elements of the stone (radius, border, markings), maintaining their proportions.

/pgfgo/stone/black (no value)

Assigns the stone the characteristics of the first player (black by default), including fill and border color. This is the player used by `\stones` unless another one is specified; `\stone`, instead, uses the neutral player.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

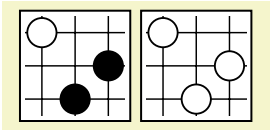
\begin{goban}
 \stones[b]{C2,D3}
 \stone[black]{B4}
\end{goban}
```

/pgfgo/stone/b (no value)

Abbreviation for ”black,” with identical functionality.

/pgfgo/stone/white (no value)

Assigns the stone the characteristics of the second player (white by default), with a white fill and black border.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[white]{B4}
\end{goban}

\begin{goban}
 \stones[white]{C2,D3}
 \stone[white]{B4}
\end{goban}
```

/pgfgo/stone/w (no value)

Abbreviation for ”white,” with the same functionality.

/pgfgo/stone/neutral (no value)

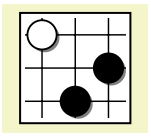
The stone belongs to the neutral player. It is the default player of `\stone`.

/pgfgo/stone/n (no value)

Abbreviation for ”neutral”.

/pgfgo/stone/drop shadow = ⟨*options*⟩ (no default)

Shadow of the stone: `true`, `false`, `angle`, `offset` and `opacity`. The options are described in the player reference (p. 22).

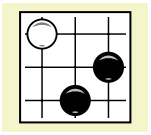


```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b, drop shadow = {true, angle = -45, offset = 1.5pt}]{C2,D3}
 \stone[w, drop shadow = {true, angle = -45, offset = 1.5pt}]{B4}
\end{goban}
```

/pgfgo/stone/drop shine = ⟨*options*⟩ (no default)

Highlight of the stone: `true`, `false`, `angle`, `opacity` and `color`. The options are described in the player reference (p. 22).

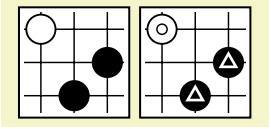


```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b, drop shine = true]{C2,D3}
 \stone[w, drop shine = {true, color = gray}]{B4}
\end{goban}
```

/pgfgo/stone/mark = ⟨*mark keys*⟩ (no default)

Places a mark on the stone (e.g., triangle, circle). Marks can be further customized with other keys, such as color or size.



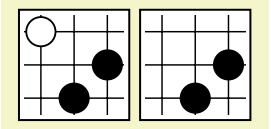
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b, mark = triangle]{C2,D3}
 \stone[w, mark = t-circle]{B4}
\end{goban}
```

/pgfgo/stone/forget (no value)

When the board uses a memory system (`new remember`, `resume`), this option indicates that the stone should not be remembered, allowing it to be excluded from subsequent states.



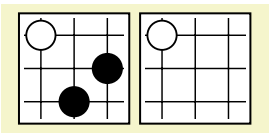
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \stones[b]{C2,D3}
 \stone[w, forget]{B4}
\end{goban}

\begin{goban}[resume]
\end{goban}
```

/pgfgo/stone/remember (no value)

In a board with active memory, ensures that the stone is remembered even if the general environment uses `forget`.



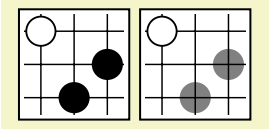
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[forget]
 \stones[b]{C2,D3}
 \stone[w, remember]{B4}
\end{goban}

\begin{goban}[resume]
\end{goban}
```

/pgfgo/stone/prisoner=⟨*prisoner options*⟩ (default `fill opacity = 0.5`)

Characterizes a stone as a prisoner, typically with reduced opacity (0.5 by default) to indicate it has been captured. Other options can be added to customize its appearance.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b, prisoner]{C2,D3}
 \stone[white]{B4}
\end{goban}
```

## 5.2 Moves

Move commands allow representing sequences of plays, automatically alternating between black and white. They include options for numbering the moves and customizing the associated stones.

\move*[⟨*options*⟩]{⟨*intersection*⟩}

Draws a stone as part of a move sequence, automatically alternating colors (black, white, black, etc.). The starred version (`\move*`) adds a numeric label to the stone.

\moves*[⟨*options*⟩]{⟨*intersection , intersection, ...*⟩}

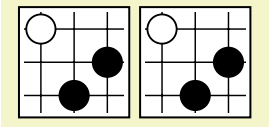
Similar to `\move`, but applies the sequence to multiple intersections in a single line.

Within the same `goban`, numbering and turn carry over from one `\moves` to the next. Each new `goban` starts at move 1 with Black; to continue a game in another diagram, use `from` and `b` or `w`, or `resume`.

It is possible to change options for all players using `\pgfgoset{move = {...}}`, which is equivalent to `\pgfgoset{players = {all = {move = {...}}}}`.

/pgfgo/move/black (no value)

Sets the move sequence to start with the black player, useful for forcing the start of a play in specific diagrams.



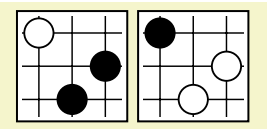
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \moves[b]{C2,B4}
 \move{D3}
\end{goban}
```

/pgfgo/move/white (no value)

Sets the sequence to start with the white player, altering the natural order of moves.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \moves[white]{C2,B4}
 \move{D3}
\end{goban}
```

/pgfgo/move/b (no value)

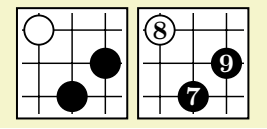
Short for `black`.

/pgfgo/move/w (no value)

Short for `white`.

/pgfgo/move/from = ⟨*number*⟩ (no default)

Defines the starting number of a numbered move sequence, allowing continuation from a specific point instead of starting at 1.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

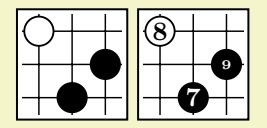
\begin{goban}
 \moves*[from = 7]{C2,B4}
 \move*{D3}
\end{goban}
```

/pgfgo/move/label = ⟨*options*⟩ (no default)

Controls the properties of numeric labels in starred moves, such as size, color, or format.

/pgfgo/move/label/font size=⟨*dimension*⟩ (no default, initially `automatic`)

Adjusts the font size of the labels, allowing them to adapt to the stone size or overall design. If it is not set, the size adapts to the number of digits.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

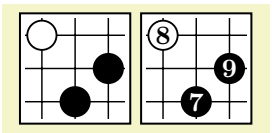
\begin{goban}
 \moves*[from = 7]{C2,B4}
 \move*[label = {font size = 0.5em}]{D3}
\end{goban}
```

/pgfgo/move/label/format = ⟨*options*⟩ (no default, initially `arabic`)

Defines the numbering style of the labels.

/pgfgo/move/label/format/arabic (no value)

Uses Arabic numerals, the default format for move labels.



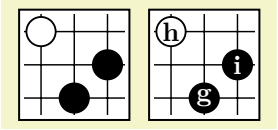
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \moves*[from = 7]{C2,B4}
 \move*{D3}
\end{goban}
```

/pgfgo/move/label/format/alph (no value)

Uses lowercase letters for the labels.



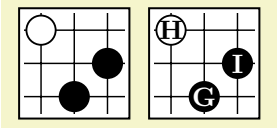
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\pgfgoset{move={label = {format=alph}}}
\begin{goban}
 \moves*[from = 7]{C2,B4}
 \move*{D3}
\end{goban}
```

/pgfgo/move/label/format/Alph (no value)

Uses uppercase letters.



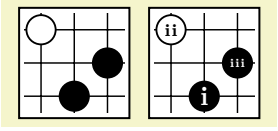
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\pgfgoset{move = {label = {format = Alph}}}
\begin{goban}
 \moves*[from = 7]{C2,B4}
 \move*{D3}
\end{goban}
```

/pgfgo/move/label/format/roman (no value)

Uses lowercase Roman numerals.



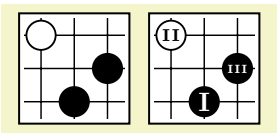
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\pgfgoset{move = {label = {format = roman}}}
\begin{goban}
 \moves*{C2,B4}
 \move*{D3}
\end{goban}
```

/pgfgo/move/label/format/Roman (no value)

Uses uppercase Roman numerals.



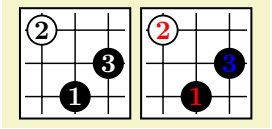
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\pgfgoset{move = {label = {format = Roman}}}
\begin{goban}
 \moves*{C2,B4}
 \move*{D3}
\end{goban}
```

/pgfgo/move/label/color = ⟨*color*⟩ (no default)

Defines the color of the labels.



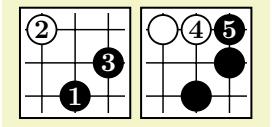
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \moves*{C2,B4}
 \move*{D3}
\end{goban}

\begin{goban}
 \moves*[label = {color = red}]{C2,B4}
 \move*[label = {color = blue}]{D3}
\end{goban}
```

/pgfgo/move/label/forget (no value)

Prevents move labels from being stored in the board's memory when using `remember`, allowing stones to persist without their numbers.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}[new remember]
 \moves*[label = forget]{C2,B4,D3}
\end{goban}

\begin{goban}[resume]
 \moves*{C:D4}
\end{goban}
```

/pgfgo/move/label/remember (no value)

Ensures that labels are stored in memory, even if the environment uses `forget`, maintaining their continuity in subsequent diagrams.

/pgfgo/move/forget (no value)

Indicates that moves should not be remembered in the board's memory, removing them from future states when using `resume`.

/pgfgo/move/remember (no value)

Ensures that moves are remembered, preserving them for reuse with `resume`.

## 5.3 SGF Games

SGF is the format that almost every Go program uses to save games. With the Lua kernel (p. 12), PGFGO can read them.

\sgf[⟨*options*⟩]{⟨*file*⟩}

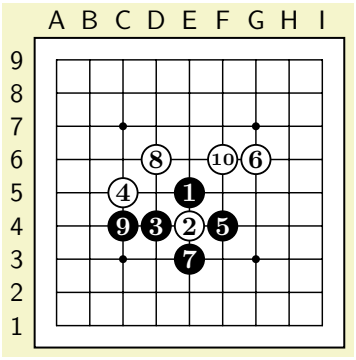
Draws on the `goban` the main line of the first game in the file: the setup stones (`AB` and `AW`) as with `\stones`, and the moves as with `\moves*`. Variations are not drawn. Passes count for the numbering, but are not drawn. It only exists inside `goban`.

The stones that the nodes after the first move add (`AB`, `AW`) or remove (`AE`) are applied together with the previous move. Removing stones needs `captures`; without `captures`, `\sgf` warns once and does not remove them.

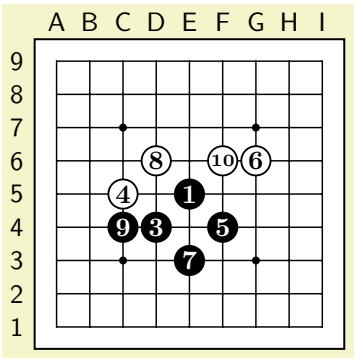
When it ends, the numbering and the turn are ready to continue the game with `\moves`. `\sgf` works with `partial`, `captures`, `remember` and `resume`. If the game is for a board of another size, a warning is produced. With `kernel = expl3`, `\sgf` raises an error.

In the examples, `game.sgf` contains a short 9×9 game:

```
(;GM[1]FF[4]SZ[9]PB[Black]PW[White]C[A short game.]
;B[ee];W[ef];B[df];W[ce];B[ff];W[gd];B[eg]C[Black captures.];W[dd];B[cf];W[fd])
```



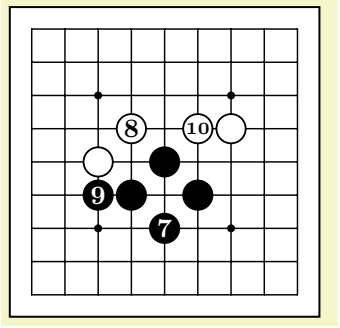

```
\begin{goban}[label = {at = {north, west}}]
 \sgf{game.sgf} % Without captures, 2 stays.
\end{goban}
```




```
\begin{goban}[label = {at = {north, west}}, captures]
 \sgf{game.sgf}
\end{goban}
```

`/pgfgo/sgf/from` = ⟨*number*⟩ (no default, initially `1`)

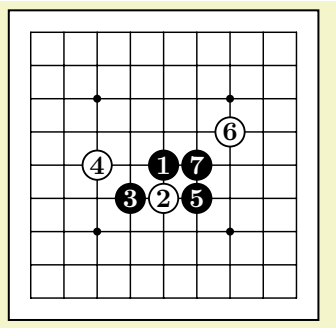
Number of the first numbered move. The moves before it are drawn without numbers.



```
\begin{goban}[captures]
 \sgf[from = 7]{game.sgf}
\end{goban}
```

`/pgfgo/sgf/to` = ⟨*number*⟩ (no default, initially `the last move`)

Number of the last move that is played. The following ones are not drawn.

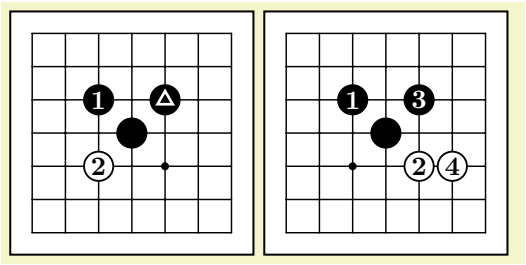


```
\begin{goban}[captures]
 \sgf[to = 6]{game.sgf}
 \moves*{F5} % Continues with 7 (Black), somewhere else.
\end{goban}
```

`from` and `to` count the moves of the file. The number that is shown can be different: the `MN` property of the file changes it (the following moves continue from there), and so does `restart`. In the following examples, `variations.sgf` is a 7×7 game with two branches, marks and `MN[10]` on the fourth move of the main line.

`/pgfgo/sgf/variation` = ⟨*numbers*⟩ (no default, initially `the main line`)

Chooses a variation: at each branch of the game, in order, the given variation is taken (1 is the main one). For example, `variation = 2.1` takes the second variation at the first branch and the first one at the second. When the list ends, the first variation is followed. If a variation does not exist, a warning is produced.

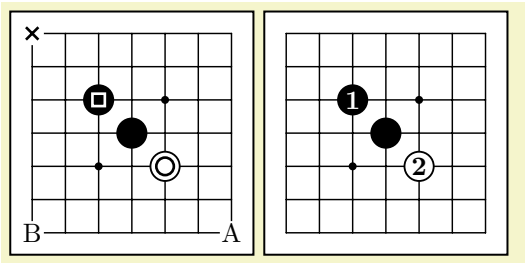


```
\begin{goban}[sgf size = variations.sgf]
 \sgf[variation = 2]{variations.sgf}
\end{goban}

\begin{goban}[sgf size = variations.sgf]
 \sgf[variation = 1.2]{variations.sgf}
\end{goban}
```

`/pgfgo/sgf/marks` =⟨*boolean*⟩ (default `true`, initially `true`)

Draws the marks of the file (`TR`, `SQ`, `CR`, `MA`, `SL` and the labels of `LB`), the arrows (`AR`) and the lines (`LN`) of the last node that is shown, and dims the points of `DD` (which stay dimmed until the next `DD`; p. 68). Each mark takes the color of the stone on that point, so that it can be seen, and replaces the number of the move. The territory of the file (`TB` and `TW`) is drawn as that of `\territory` (p. 42) and counts for `\pgfgoscore`.

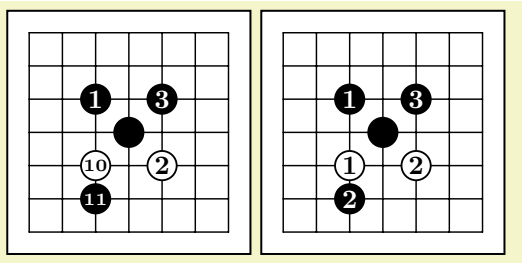


```
\begin{goban}[sgf size = variations.sgf]
 \sgf[to = 2]{variations.sgf}
\end{goban}

\begin{goban}[sgf size = variations.sgf]
 \sgf[to = 2, marks = false]{variations.sgf}
\end{goban}
```

/pgfgo/sgf/restart = ⟨*number*⟩ (no default, initially 0)

Restarts the numbering every ⟨*number*⟩ moves, as books do: with `restart = 100`, move 101 is shown as 1. With 0, it does not restart.




```
\begin{goban}[sgf size = variations.sgf]
 \sgf[]{variations.sgf}
\end{goban}

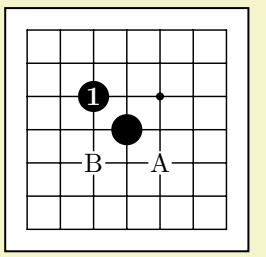
\begin{goban}[sgf size = variations.sgf]
 \sgf[restart = 3]{variations.sgf}
\end{goban}
```

/pgfgo/sgf/game = ⟨*number*⟩ (no default, initially 1)

An SGF file can have several games (a collection). It chooses which one is read; if it does not exist, the first one is read with a warning. It also applies to `\sgfanimation`, `\sgfdiagrams`, `\sgfcomment`, `\sgfinfo` and `sgf size`, which have their own `game` *key*; for the whole document, `\pgfgoset{sgf = {game = 2}}`.

/pgfgo/sgf/variations =⟨*none, children, siblings or auto*⟩ (default `auto`, initially `none`)

Marks with letters the variations of the last move that is shown: with `children`, the moves that can follow; with `siblings`, the alternatives to that move (the letter replaces its number). `auto` does what the file asks for with `ST`. They are only marked if there is more than one.




```
\begin{goban}[sgf size = variations.sgf]
 \sgf[to = 1, marks = false, variations = children]{variations.sgf}
\end{goban}
```

`\sgf` also reads `PL` (who plays next: `\moves` goes on with that color) and `CA` (the charset of the file: Latin-1 files are converted; other charsets produce a warning).

/pgfgo/sgf = ⟨*sgf options*⟩ (no default)

Options of every `\sgf` in the document, for example `\pgfgoset{sgf = {variations}}`.

`\sgfinfo[`⟨*options*⟩`]{`⟨*file*⟩`}{`⟨*property*⟩`}`

Writes a property of the root of the game, for example `PB` and `PW` (the players), `RE` (the result), `KM` (the komi) or `DT` (the date). If it is missing, it writes nothing. It can be used outside `goban`.

Ann (Black) against Bob (White), komi 6.5. Result: B+R.

```
\sgfinfo{variations.sgf}{PB} (Black) against \sgfinfo{variations.sgf}{PW} (White), komi \sgfinfo{variations.sgf}{KM}. R
```

`\sgfcomment[`⟨*options*⟩`]{`⟨*file*⟩`}{`⟨*move*⟩`}`

Writes the comment of a move of the file (`0`: the one before the first move): the name of the node (`N`), in bold; the annotations of the move (`TE`, `BM`, `DO`, `IT`) and of the position (`GB`, `GW`, `DM`, `UC`); and the comment (`C`). The annotations are written with the texts of `/pgfgo/sgf annotations`. Values (`V`) and hotspots (`HO`), which are meant for the game tree, are not written. It can be used outside `goban`. `\sgfanimation` and `\sgfdiagrams` write the comments in the same way.

Move 7: Tesuji. Black captures.

```
Move 7: \sgfcomment{game.sgf}{7}
```

`/pgfgo/sgf info/game` = ⟨*number*⟩ (no default, initially `1`)

Like `/pgfgo/sgf/game`, for `\sgfinfo`.

`/pgfgo/sgf comment/game` = ⟨*number*⟩ (no default, initially `1`)

Like `/pgfgo/sgf/game`.

`/pgfgo/sgf comment/variation` = ⟨*numbers*⟩ (no default, initially `the main line`)

Like `/pgfgo/sgf/variation`.

`/pgfgo/sgf comment/annotations` =⟨*boolean*⟩ (default `true`, initially `true`)

Writes the name and the annotations. With `false`, only the comment.

The texts of the annotations are in English, and they can be changed for the whole document with `\pgfgoset{sgf annotations = {...}}`, for example `bad move = Mala jugada`. The emphasized annotations of `DM` and `UC` use the same text.

`/pgfgo/sgf annotations/tesuji` = ⟨*text*⟩ (no default, initially `Tesuji`)

Text of `TE`.

`/pgfgo/sgf annotations/very good tesuji` = ⟨*text*⟩ (no default, initially `Very good tesuji`)

Text of `TE[2]`.

`/pgfgo/sgf annotations/bad move` = ⟨*text*⟩ (no default, initially `Bad move`)

Text of `BM`.

`/pgfgo/sgf annotations/very bad move` = ⟨*text*⟩ (no default, initially `Very bad move`)

Text of `BM[2]`.

`/pgfgo/sgf annotations/doubtful` = ⟨*text*⟩ (no default, initially `Doubtful move`)

Text of `DO`.

`/pgfgo/sgf annotations/interesting` = ⟨*text*⟩ (no default, initially `Interesting move`)

Text of `IT`.

`/pgfgo/sgf annotations/good for black` = ⟨*text*⟩ (no default, initially `Good for Black`)

Text of `GB`.

`/pgfgo/sgf annotations/very good for black` = ⟨*text*⟩ (no default, initially `Very good for Black`)

Text of `GB[2]`.

`/pgfgo/sgf annotations/good for white` = ⟨*text*⟩ (no default, initially `Good for White`)

Text of `GW`.

`/pgfgo/sgf annotations/very good for white` = ⟨*text*⟩ (no default, initially `Very good for White`)

Text of `GW[2]`.

`/pgfgo/sgf annotations/even` = ⟨*text*⟩ (no default, initially `Even position`)

Text of `DM`.

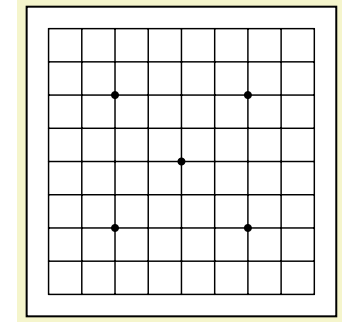
`/pgfgo/sgf annotations/unclear` = ⟨*text*⟩ (no default, initially `Unclear position`)

Text of `UC`.

`\sgfanimation[`⟨*options*⟩`]{`⟨*file*⟩`}`

Steps through the game with the `animate` package: it draws one `goban` per move, and the `animate` buttons go forward, back or play it. Frame $k$ shows the position after move $k$; frame 0, the position before the first move. It is used outside `goban`. The boards have the size of the game (`SZ`), unless `goban` gives another one. It needs `\usepackage{animate}` and two runs.

`animate` animations work in Acrobat Reader and in a few other viewers (Foxit, PDF-XChange). Other viewers only show the first frame. These options can be set for the whole document with `\pgfgoset{sgf animation = {...}}`.



A short game.

```
\sgfanimation[comments, goban = {captures}]{game.sgf}
```

`/pgfgo/sgf animation/from` = ⟨*number*⟩ (no default, initially `0`)

First frame.

`/pgfgo/sgf animation/to` = ⟨*number*⟩ (no default, initially `the last move`)

Last frame.

`/pgfgo/sgf animation/fps` = ⟨*number*⟩ (no default, initially `1`)

Frames per second when playing.

`/pgfgo/sgf animation/goban` = ⟨*goban options*⟩ (no default)

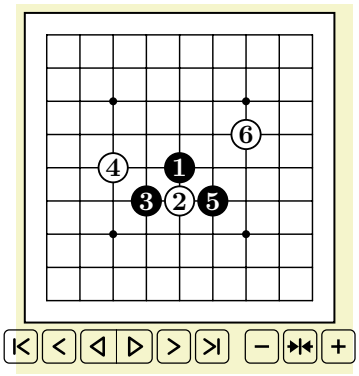
Options of every board, as in `\begin{goban}[...]`.

`/pgfgo/sgf animation/animate` = ⟨*animate options*⟩ (no default, initially `controls`)

Options of the `animateinline` environment of `animate`, for example `controls, loop`.

/pgfgo/sgf animation/numbers =⟨*boolean*⟩ (default `true`, initially `false`)

Numbers every move, as `\sgf` does. Without `numbers`, moves have no numbers and the last one is marked.



```
\sgfanimation[from = 6, to = 8, numbers, goban = {captures}]{game.sgf}
```

/pgfgo/sgf animation/last mark = ⟨*mark options or false*⟩ (no default, initially `circle`)

Mark of the last move, in the colour of the player who made it. With `false`, the last move is not marked. It is not used with `numbers`, nor when the file already marks that point.

/pgfgo/sgf animation/comments =⟨*boolean*⟩ (default `true`, initially `false`)

Shows the comment (`C`) of the move below the board. The one of frame 0 is the comment of the nodes before the first move.

/pgfgo/sgf animation/comment height = ⟨*dimension*⟩ (no default, initially `4em`)

Height of the space for the comments. It is the same in every frame, so that frames do not change size.

/pgfgo/sgf animation/comment font = ⟨*font commands*⟩ (no default, initially `\small`)

Font of the comments.

/pgfgo/sgf animation/variation = ⟨*numbers*⟩ (no default, initially `the main line`)

Like `/pgfgo/sgf/variation`.

/pgfgo/sgf animation/marks =⟨*boolean*⟩ (default `true`, initially `true`)

Like `/pgfgo/sgf/marks`: each frame shows the marks of its move.

/pgfgo/sgf animation/restart = ⟨*number*⟩ (no default, initially `0`)

Like `/pgfgo/sgf/restart`.

/pgfgo/sgf animation/annotations =⟨*boolean*⟩ (default `true`, initially `true`)

With `comments`, it also writes the name and the annotations of each move, like `\sgfcomment`.

/pgfgo/sgf animation/game = ⟨*number*⟩ (no default, initially `1`)

Like `/pgfgo/sgf/game`.

/pgfgo/sgf animation/view =⟨*boolean*⟩ (default `true`, initially `true`)

If the file has a view (`VW`), the boards only show that part, like `sgf view`.

/pgfgo/sgf animation/layers =⟨*boolean*⟩ (default `true`, initially `true`)

Builds the animation by layers: the board is drawn only once, and so is every stone, on a transparency that `animate` shows while the stone stays on the board (with the `timeline` option of `animate`). The only things drawn in every frame are the things that change: the mark of the last move, the marks of the file and the comment. So the PDF grows with the number of moves and not with its square: a 200-move game takes less than 1 MB instead of more than 10 MB, and it also compiles faster. With `layers = false`, every frame is a whole `goban`. The `timeline` is saved in an auxiliary file, `<jobname>-pgfgo-<n>.tln`.

`\sgfdiagrams[`⟨*options*⟩`]{`⟨*file*⟩`}`

Draws the game as books do: one diagram every `step` moves (1–50, 51–100, etc.). Each diagram starts from the real position, with the earlier moves without numbers and their captures; stones captured within the diagram are not removed (they are removed in the next one). Below each one go its caption and its "N at M" notes (p. 45). It is used outside `goban`. The options can be set for the whole document with `\pgfgoset{sgf diagrams = {...}}`.

```
\sgfdiagrams[step = 5]{game.sgf}
```

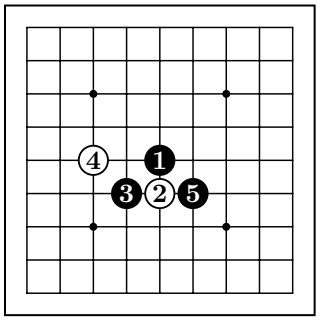



Diagram 1 (1–5)

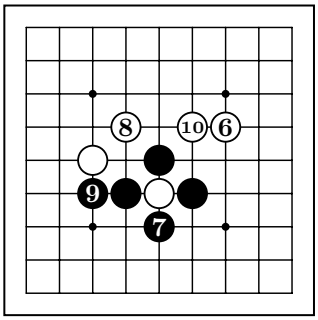



Diagram 2 (6–10)

`/pgfgo/sgf diagrams/step =` ⟨*number*⟩ (no default, initially `50`)

Number of moves of each diagram.

`/pgfgo/sgf diagrams/to =` ⟨*number*⟩ (no default, initially `the last move`)

Last move.

`/pgfgo/sgf diagrams/restart =`⟨*boolean*⟩ (default `true`, initially `false`)

Each diagram is numbered from 1, as some books do.

`/pgfgo/sgf diagrams/notes =`⟨*boolean*⟩ (default `true`, initially `true`)

Writes the "N at M" notes below each diagram.

`/pgfgo/sgf diagrams/title =` ⟨*text*⟩ (no default, initially `Diagram`)

Word of the caption: "Diagram 2 (51–100)".

`/pgfgo/sgf diagrams/caption font =` ⟨*font commands*⟩ (no default, initially `\small`)

Font of the caption and the notes.

`/pgfgo/sgf diagrams/goban =` ⟨*goban options*⟩ (no default)

Options of every board. By default, the boards have the size of the game and use `captures` and `notes`.

`/pgfgo/sgf diagrams/variation =` ⟨*numbers*⟩ (no default, initially `the main line`)

Like `/pgfgo/sgf/variation`.

`/pgfgo/sgf diagrams/comments =`⟨*boolean*⟩ (default `true`, initially `false`)

Writes below each diagram the comments of its moves, like `\sgfcomment`, each one with the number of the move.

`/pgfgo/sgf diagrams/annotations =`⟨*boolean*⟩ (default `true`, initially `true`)

With `comments`, it also writes the name and the annotations of each move.

`/pgfgo/sgf diagrams/figures =`⟨*boolean*⟩ (default `true`, initially `true`)

If the file marks figures (`FG`), each figure is a diagram, with its name as caption, instead of one every `step` moves.

`/pgfgo/sgf diagrams/view =`⟨*boolean*⟩ (default `true`, initially `true`)

Like `/pgfgo/sgf animation/view`.

`/pgfgo/sgf diagrams/game =` ⟨*number*⟩ (no default, initially `1`)

Like `/pgfgo/sgf/game`.

### 5.4 Prisoners

PGFGO provides specific commands to represent prisoners, typically with reduced opacity to distinguish them from active stones.

`\prisoner[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

Draws a prisoner stone at a specific intersection, with options to customize its style.

`\prisoners[`⟨*options*⟩`]{`⟨*intersection , intersection, ...*⟩`}`

Places multiple prisoner stones in a single line, efficient for showing captured groups.

`/pgfgo/prisoner=`⟨*stone options*⟩ (default `fill opacity = 0.5`)

Defines the characteristics of prisoner stones, with a default fill opacity of 0.5 to indicate their status. Other `\stone` options, such as color or size, can be applied.

## 6 Marks

Marks are graphical elements placed at the intersections of the board to highlight specific positions, indicate points of interest, or number sequences without the need to add stones. PGFGO offers versatile commands and options to customize these marks, ranging from predefined shapes to custom labels.

`\mark[`⟨*options*⟩`]{`⟨*intersection*⟩`}`

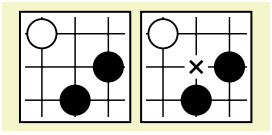
Draws an individual mark at the specified intersection (e.g., "C3"). Options allow defining its shape, color, and other attributes.

`\marks[`⟨*options*⟩`]{`⟨*intersection , intersection, ...*⟩`}`

Similar to `\mark`, but applies the same mark to multiple intersections using the coordinate syntax.

`/pgfgo/mark =`⟨*options*⟩ (default `cross`)

Defines the type of mark to be placed, with a cross ("cross") as the default value if no other option is specified.



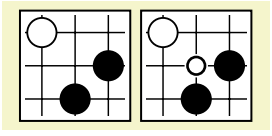
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark{C3}
\end{goban}
```

`/pgfgo/mark/circle` (no value)

Places a hollow circle as a mark.



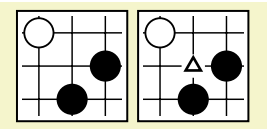
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[circle]{C3}
\end{goban}
```

/pgfgo/mark/triangle (no value)

Draws a hollow triangle.



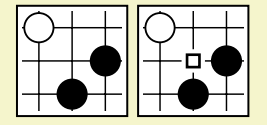
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[triangle]{C3}
\end{goban}
```

/pgfgo/mark/square (no value)

Places a hollow square.



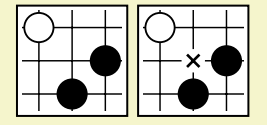
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[square]{C3}
\end{goban}
```

/pgfgo/mark/cross (no value)

Draws a cross.



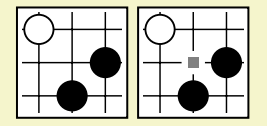
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[cross]{C3}
\end{goban}
```

/pgfgo/mark/filled square (no value)

Places a filled square.



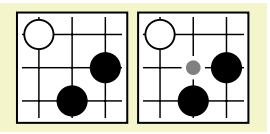
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[filled square]{C3}
\end{goban}
```

/pgfgo/mark/filled circle (no value)

Draws a filled circle.



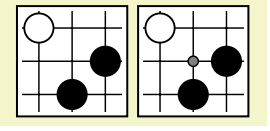
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[filled circle]{C3}
\end{goban}
```

/pgfgo/mark/t-circle (no value)

Places a small circle with a thicker border, specifically designed for marking territories. It can also be written `tcircle`.



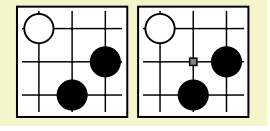
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[t-circle]{C3}
\end{goban}
```

/pgfgo/mark/t-square (no value)

Draws a small square with a thick border, similar to ”t-circle,” used to mark territories. It can also be written `tsquare`.



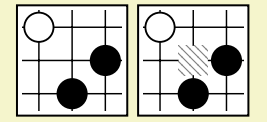
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[t-square]{C3}
\end{goban}
```

/pgfgo/mark/hatched lines (no value)

Places a pattern of crossed (hatched) lines.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[hatched lines]{C3}
\end{goban}
```

/pgfgo/mark/sequence=⟨*options*⟩ (default `from = 1`)

Allows numbering a series of marks in order.

/pgfgo/mark/sequence/use (no value)

Displays the `sequence` mark on the board. It is the same as `sequence = true`.

/pgfgo/mark/sequence/false (no value)

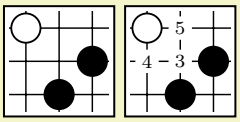
Turns the sequence off (for example, one turned on with `\pgfgoset`).

/pgfgo/mark/sequence/format = ⟨*format*⟩ (no default)

Turns the sequence on and chooses the number format. It is the same as `sequence, label = {format = ...}` (see `label/format`).

/pgfgo/mark/sequence/from=⟨*number*⟩ (no default)

Sets the starting number of a numbered mark sequence, allowing continuation from a specific point instead of 1.



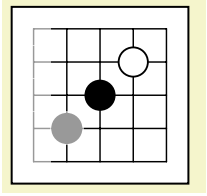

```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[sequence = {use, from = 3}]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/dim (no value)

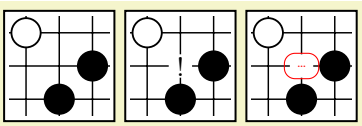
Dims the point: it covers it with the color of the background of the board, translucent, over the stone and the grid. It is what `\sgf` draws for the `DD` property.



```
\begin{goban}[5]
 \stones{B2, C3}\stone[w]{D4}
 \marks[dim]{A1:5, B2}
\end{goban}
```

/pgfgo/mark/custom=⟨*token list*⟩ (no default)

Allows defining a custom mark using text or symbols (e.g., "!" or "?") and even images.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[custom = {!}]{C3}
\end{goban}

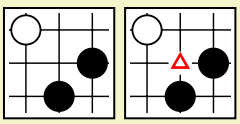
\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[custom = {\tikz{\node[rounded corners, draw, inner sep = 1em, red]{\dots};}}]{C3}
\end{goban}
```

/pgfgo/mark/line = ⟨*options*⟩ (no default)

Controls the outline of the mark.

/pgfgo/mark/line/color = ⟨*color*⟩ (no default)

Defines the outline color of the mark.



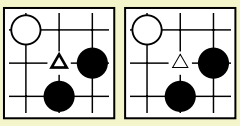
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[triangle, line = {color = red}]{C3}
\end{goban}
```

/pgfgo/mark/line/width = ⟨*dimension*⟩ (no default)

Adjusts the thickness of the mark's outline.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[triangle]{C3}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[triangle, line = {width = 0.1pt}]{C3}
\end{goban}
```

/pgfgo/mark/fill = ⟨*options*⟩ (no default)

Controls the fill of the filled marks (`filled square`, `filled circle` and `hatched lines`).

/pgfgo/mark/fill/true (no value)

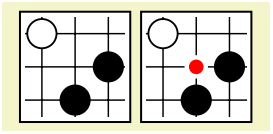
Filled marks use `fill/color`. Without `fill = true`, they are filled in gray.

/pgfgo/mark/fill/false (no value)

Filled marks are filled in gray again. It is the initial value.

/pgfgo/mark/fill/color = ⟨*color*⟩ (no default)

Sets the fill color of a mark. It is only used with `fill = true`.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

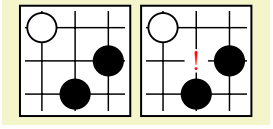
\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[filled circle, fill = {true, color = red}]{C3}
\end{goban}
```

/pgfgo/mark/label = ⟨*option*⟩ (no default)

Controls the appearance of labels for marks that have them.

/pgfgo/mark/label/color = ⟨*color*⟩ (no default)

Controls the color of the text or symbol in custom marks (e.g., with `custom`).



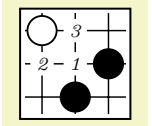
```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[custom = {!}, label = {color = red}]{C3}
\end{goban}
```

/pgfgo/mark/label/font = ⟨*font commands*⟩ (no default)

Font of the labels of marks (`sequence` and `custom`).




```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

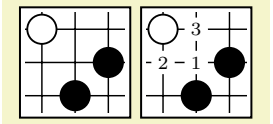
\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[sequence, label = {font = \itshape}]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/label/format=⟨*options*⟩ (no default, initially `arabic`)

Controls the numbering style of enumeration labels (e.g., `sequence`), with options such as Arabic numerals, letters, or Roman numerals.

/pgfgo/mark/label/format/arabic (no value)

Uses Arabic numerals for sequences.



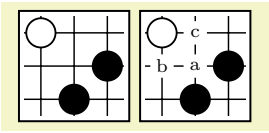

```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[label = {format = arabic}, sequence]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/label/format/alph (no value)

Uses lowercase letters for the sequence.



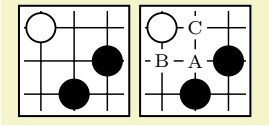

```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[label = {format = alph}, sequence]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/label/format/Alph (no value)

Uses uppercase letters.



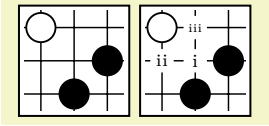

```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[label = {format = Alph}, sequence]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/label/format/roman (no value)

Uses lowercase Roman numerals.



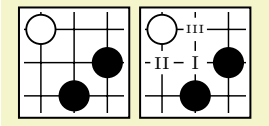

```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[label = {format = roman}, sequence]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/label/format/Roman (no value)

Uses uppercase Roman numerals.



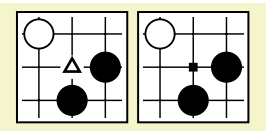

```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \marks[label = {format = Roman}, sequence]{C3,B3,C4}
\end{goban}
```

/pgfgo/mark/black (no value)

Applies the properties of the first player (black) to the mark.



```
\pgfgoset{goban = {5, partial = {from = B2 to D4}}}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[triangle]{C3}
\end{goban}

\begin{goban}
 \stones[b]{C2,D3}
 \stone[w]{B4}
 \mark[black, t-square]{C3}
\end{goban}
```

/pgfgo/mark/white (no value)

Applies the properties of the second player (white) to the mark.

/pgfgo/mark/neutral (no value)

Keeps the mark unassociated with any player.

/pgfgo/mark/b (no value)

Short for `black`.

/pgfgo/mark/w (no value)

Short for `white`.

/pgfgo/mark/n (no value)

Short for `neutral`.

/pgfgo/mark/forget (no value)

This mark is not stored in the board's memory, even if the `goban` uses `remember`.

/pgfgo/mark/remember (no value)

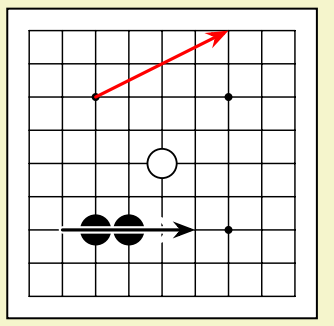
This mark is stored in the board's memory, even if the `goban` uses `forget`.

## 6.1 Arrows and Lines

Arrows and lines join two intersections. They are drawn above the stones, with a halo in the color of the background of the board that separates them from the stones and the grid. Like `\mark`, `\arrow` and `\line` only exist inside `goban`: outside, `\line` is still the LaTeX command. `\sgf` draws the arrows (`AR`) and lines (`LN`) of the file.

`\arrow[`⟨*options*⟩`]{`⟨*from*⟩`}{`⟨*to*⟩`}`

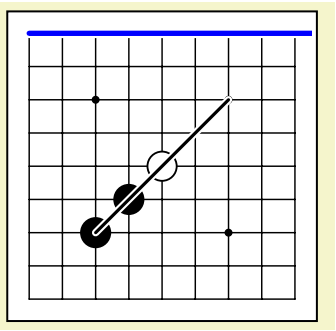
Draws an arrow from one intersection to another.



```
\begin{goban}[9]
 \stones{C3, D3}\stone[w]{E5}
 \arrow{B3}{F3}
 \arrow[color = red, halo = false]{C7}{G9}
\end{goban}
```

`\line[`⟨*options*⟩`]{`⟨*from*⟩`}{`⟨*to*⟩`}`

Draws a line from one intersection to another.



```
\begin{goban}[9]
 \stones{C3, D4}\stone[w]{E5}
 \line{C3}{G7}
 \line[color = blue, width = 2pt]{A9}{J9}
\end{goban}
```

`/pgfgo/arrow/color=`⟨*color*⟩ (no default, initially `black`)

Color of the arrow.

`/pgfgo/arrow/width=`⟨*dimension*⟩ (no default, initially `1.2pt`)

Width of the arrow. The tip grows with it.

`/pgfgo/arrow/opacity=`⟨*number*⟩ (no default, initially `1`)

Opacity of the arrow.

`/pgfgo/arrow/tip=`⟨*arrow tip*⟩ (no default, initially `Stealth`)

The tip of the arrow, one of the tips of the `arrows.meta` library of Ti*k*Z, like `Latex` or `Triangle`.

`/pgfgo/arrow/halo=`⟨*halo options*⟩ (default `true`)

The halo around the arrow.

`/pgfgo/arrow/halo/true` (style, no value)

Draws the halo. It is the initial value.

`/pgfgo/arrow/halo/false` (style, no value)

Does not draw the halo.

`/pgfgo/arrow/halo/color=`⟨*color*⟩ (no default)

Color of the halo. Without this *key*, it is the color of the background of the board.

`/pgfgo/arrow/halo/width=`⟨*dimension*⟩ (no default, initially `0.8pt`)

Width of the halo on each side of the arrow.

`/pgfgo/arrow/remember` (no value)

This arrow is saved in the memory of the board, even if the `goban` uses `forget`.

/pgfgo/arrow/forget (no value)

This arrow is not saved in the memory of the board, even if the `goban` uses `remember`.

Lines have the same options, except `tip`:

/pgfgo/line/color=⟨*color*⟩ (no default, initially `black`)

Like `/pgfgo/arrow/color`.

/pgfgo/line/width=⟨*dimension*⟩ (no default, initially `1.2pt`)

Like `/pgfgo/arrow/width`.

/pgfgo/line/opacity=⟨*number*⟩ (no default, initially `1`)

Like `/pgfgo/arrow/opacity`.

/pgfgo/line/halo=⟨*halo options*⟩ (default `true`)

Like `/pgfgo/arrow/halo`.

/pgfgo/line/halo/true (style, no value)

Like `/pgfgo/arrow/halo/true`.

/pgfgo/line/halo/false (style, no value)

Like `/pgfgo/arrow/halo/false`.

/pgfgo/line/halo/color=⟨*color*⟩ (no default)

Like `/pgfgo/arrow/halo/color`.

/pgfgo/line/halo/width=⟨*dimension*⟩ (no default, initially `0.8pt`)

Like `/pgfgo/arrow/halo/width`.

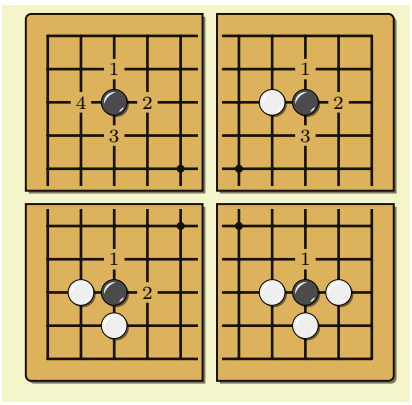
/pgfgo/line/remember (no value)

Like `/pgfgo/arrow/remember`.

/pgfgo/line/forget (no value)

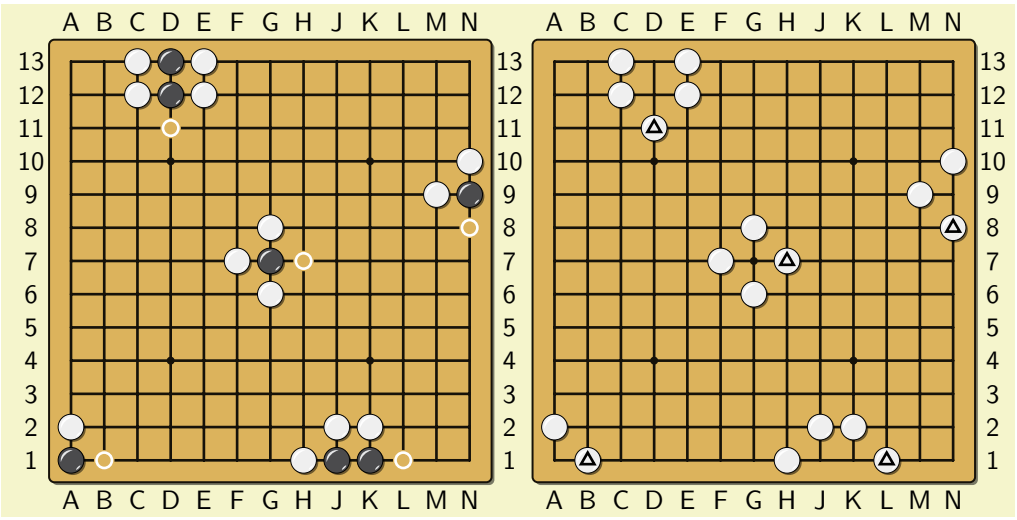
Like `/pgfgo/arrow/forget`.

# 7 Other Examples



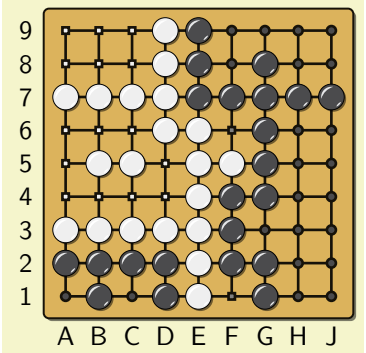
```
\setlength{\tabcolsep}{2pt}

\begin{tabular}{cc}
 \begin{goban}[partial = {from = A9 to E5}]
  \stone[b]{C7}
  \marks[sequence]{C8, D7, C6, B7}
 \end{goban} &
 %
 \begin{goban}[partial = {from = E9 to I5}]
  \stone[b]{G7}
  \stone[w]{F7}
  \marks[sequence]{G8, H7, G6}
 \end{goban} \\
 %
 \begin{goban}[partial = {from = A1 to E5}]
  \stone[b]{C3}
  \stones[w]{B3,C2}
  \marks[sequence]{C4, D3}
 \end{goban} &
 %
 \begin{goban}[partial = {from = E1 to I5}]
  \stone[b]{G3}
  \stones[w]{F3,G2, H3}
  \marks[sequence]{G4}
 \end{goban}\\
\end{tabular}
```



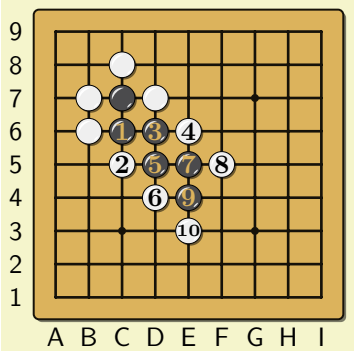

```
\pgfgoset{use i = false}

\begin{goban}[13, label = {at = {all}}]
 \stones[b]{A1, J1, K1, G7, N9, D12, D13}
 \stones[w]{A2, H1, J2, K2, G6, F7, G8, M9, N10, C12, C13, E12, E13}
 \marks[circle, line = {color = white}]{B1, L1, N8, H7, D11}
\end{goban}
\begin{goban}[13, label = {at = {north, east, south}}]
 \stones[w]{A2, H1, J2, K2, G6, F7, G8, M9, N10, C12, C13, E12, E13}
 \stones[w,mark = triangle]{B1, L1, N8, H7, D11}
\end{goban}
```

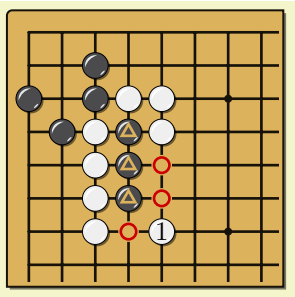



```
\pgfgoset{use i = false}

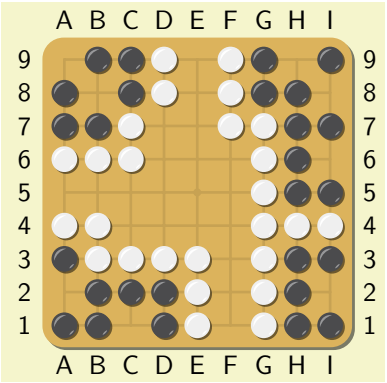
\begin{goban}[9, label = {at = {south, west}}]
 \stones[b]{A2-|D1, B1, G1|-F2|-G4|-J7-|E9, G8}
 \stones[w]{E1|-D6|-A7, D 8:9, F5, B:C 5, A:D 3}
 \marks[b, t-circle]{J1 R H6, G3, F8 R J9; G8; A1, C1}
 \marks[b, t-square, white]{A9 R C8, A4 R D6; B:C 5, D6}
 \marks[t-square, neutral]{F1,F6}
\end{goban}
```




```
\begin{goban}[9, label = {at = {south, west}}]
 \stones[b]{C7}
 \stones[w]{B6,B7,C8,D7}
 \moves*{C6,C5,D6,E6,D5,D4,E5,F5,E4,E3}
\end{goban}
```



```
\begin{goban}[9, partial = {from = A2 to H9}]
 \stones[b]{A7,B6,C8,C7}
 \stones[b, mark = triangle]{D6,D5,D4}
 \stones[w]{C3,C4,C5,C6,D7,E7,E6}
 \stone[w, mark = {custom = 1}]{E3}
 \marks[b, circle, line = {color = red!80!black}] {D3,E4, E5}
\end{goban}
```



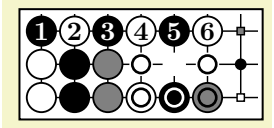

```
\pgfgouseprofile{pgfgosmooth}

\begin{goban}[label = {at = all}]
 \stones[b]{A8 |- B7, B9 -| C8, H:I 1:9, G8:9, A:D 1:2, A3; I2, I6, I8, H9, C1, A2, H:J4;}
 \stones[w]{E1 |- B3 |- A4, A6 -| C7 -| D9, G1 |- I4 -| G7 -| F9; D7;}
\end{goban}
```

# 8 Catalog

## 8.1 Profiles

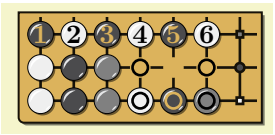
### 8.1.1 default



```
\pgfgouseprofile{default}

\begin{goban}[partial = {from = A1 to G3}]
 \moves*[from = 1]{A:F 3}
 \stones[w]{A1:2}
 \stones[b]{B1:2}
 \stones[n]{C1:2}
 \stone[w, mark = circle]{D1}
 \mark[w, circle]{D2}
 \stone[b, mark = circle]{E1}
 \mark[b, circle]{E2}
 \stone[n, mark = circle]{F1}
 \mark[n, circle]{F2}
 \mark[w, tsquare]{G1}
 \mark[b, tcircle]{G2}
 \mark[n, tsquare]{G3}
\end{goban}
```

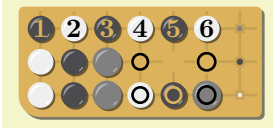
### 8.1.2 pgfgo



```
\pgfgouseprofile{pgfgo}

\begin{goban}[partial = {from = A1 to G3}]
 \moves*[from = 1]{A:F 3}
 \stones[w]{A1:2}
 \stones[b]{B1:2}
 \stones[n]{C1:2}
 \stone[w, mark = circle]{D1}
 \mark[w, circle]{D2}
 \stone[b, mark = circle]{E1}
 \mark[b, circle]{E2}
 \stone[n, mark = circle]{F1}
 \mark[n, circle]{F2}
 \mark[w, tsquare]{G1}
 \mark[b, tcircle]{G2}
 \mark[n, tsquare]{G3}
\end{goban}
```

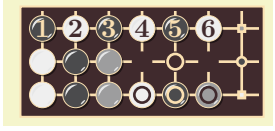
### 8.1.3 pgfgosmooth



```
\pgfgouseprofile{pgfgosmooth}

\begin{goban}[partial = {from = A1 to G3}]
 \moves*[from = 1]{A:F 3}
 \stones[w]{A1:2}
 \stones[b]{B1:2}
 \stones[n]{C1:2}
 \stone[w, mark = circle]{D1}
 \mark[w, circle]{D2}
 \stone[b, mark = circle]{E1}
 \mark[b, circle]{E2}
 \stone[n, mark = circle]{F1}
 \mark[n, circle]{F2}
 \mark[w, tsquare]{G1}
 \mark[b, tcircle]{G2}
 \mark[n, tsquare]{G3}
\end{goban}
```

### 8.1.4 wain



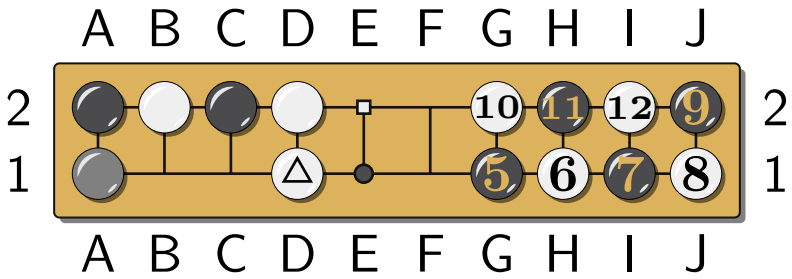
```
\pgfgouseprofile{wain}

\begin{goban}[partial = {from = A1 to G3}]
 \moves*[from = 1]{A:F 3}
 \stones[w]{A1:2}
 \stones[b]{B1:2}
 \stones[n]{C1:2}
 \stone[w, mark = circle]{D1}
 \mark[w, circle]{D2}
 \stone[b, mark = circle]{E1}
 \mark[b, circle]{E2}
 \stone[n, mark = circle]{F1}
 \mark[n, circle]{F2}
 \mark[w, tsquare]{G1}
 \mark[b, tcircle]{G2}
 \mark[n, tsquare]{G3}
\end{goban}
```

# 9 Definitions of Stones and Boards

## 9.1 Stones and Players

### 9.1.1 pgfgo




We declare all players and start by defining the appearance of the neutral one.

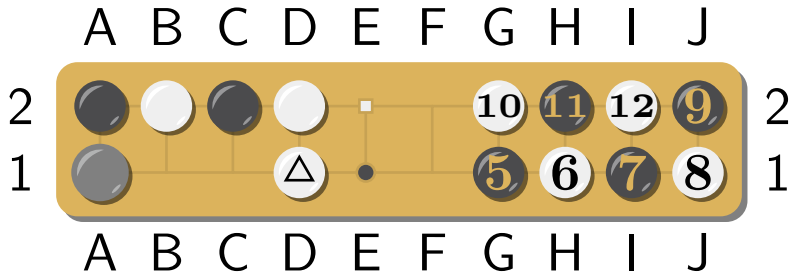
```
\pgfgodefineplayer{pgfgo-neutral}{
 stone = {
  radius = 0.19cm,
  line = {
   true,
   color = black,
   width = 0.3pt,
   opacity = 1,
  },
  fill = {
   true,
   color = gray,
   opacity = 1,
  },
  radius = 0.17cm,
  scale = 1,
  drop shadow = {true, opacity = 0.5, offset = 1pt, angle = -45},
  drop shine = {true, angle = 45, color = wainshade}
 },
 move = {
  label = {
   color = black,
   format = arabic,
  }
 },
 mark = {
  line = {
   color = black,
   width = 1pt,
  },
  fill = {false},
  label = {
   color = black,
   format = arabic
  }
 }
}
```

Some properties such as line thickness, opacity, and others should be preserved for all stones, so we copy the neutral player template.

```
\pgfgodefineplayer[from = pgfgo-neutral]{pgfgo-white}{
 stone = {
  fill = {color = wainwhite},
  drop shine = {color = white}
 },
 move = {
  label = {
   color = black
  }
 },
 mark = {
  line  = {color = black},
  fill  = {color = white},
  label = {color = black}
 }
}

\pgfgodefineplayer[from = pgfgo-neutral]{pgfgo-black}{
 stone = {
  fill = {
   color = wainblack
  }
 },
 move = {
  label = {
   color = pgfgo
  }
 },
 mark = {
  line = {
   color = pgfgo,
  },
  label = {color = black}
 }
}
```

### 9.1.2 pgfgosmooth

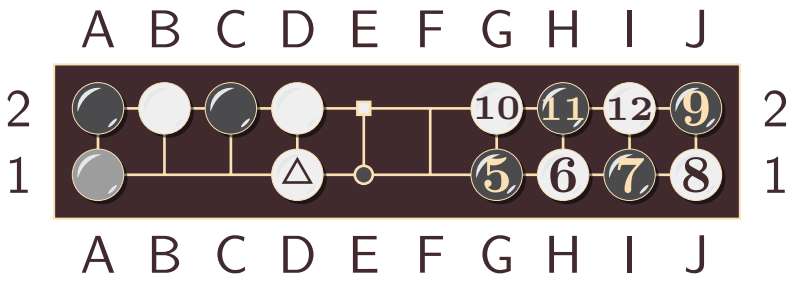



```
\pgfgodefineplayer[from = pgfgo-neutral]{pgfgosmooth-neutral}{
 stone = {
  line = false,
  radius = 0.19cm,
  drop shine = {opacity = 0.5}}
}
```

```
\pgfgodefineplayer[from = pgfgo-black]{pgfgosmooth-black}{
 stone = {line = false, drop shine = {opacity = 0.5}}
}
```

```
\pgfgodefineplayer[from = pgfgo-white]{pgfgosmooth-white}{
 stone = {line = false}
}
```

### 9.1.3 wain




```
\definecolor{waingoban}{HTML}{402A2D}
\definecolor{waingoban-s}{HTML}{493739}
\definecolor{waingrid}{HTML}{FAE0B2}
\definecolor{wainblack}{HTML}{4B4B4D} %BLACK STONE
\definecolor{wainshade}{HTML}{E6E7E8}
\definecolor{wainwhite}{HTML}{EFEFEF} %WHITE STONE
```

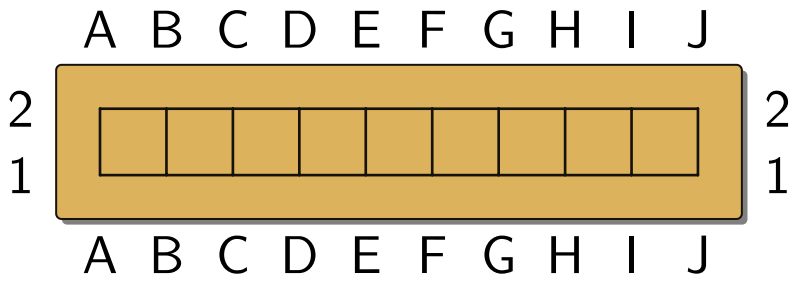
```
\pgfgodefineplayer[from = default-neutral]{wain-neutral}{
 stone = {
  line = {
   true,
   color = waingrid,
   width = 0.2pt,
   opacity = 1,
  },
  fill = {
   true,
   color = wainblack!50!wainwhite,
   opacity = 1,
  },
  radius = 0.17cm,
  scale = 1,
  drop shadow = {true, opacity = 0.5, offset = 1pt, angle = -45},
  drop shine = {true, angle = 45, color = wainshade}
 },
 move = {
  label = {
   color = black,
   format = arabic,
  }
 },
 mark = {
  line = {
   color = waingoban,
   width = 1pt,
  },
  fill = {false},
  label = {
   color = waingoban,
   format = arabic
  },
 }
}
```

```
\pgfgodefineplayer[from = wain-neutral]{wain-white}{
 stone = {
  fill = {
   color = wainwhite
  }
 },
 move = {
  label = {
   color = waingoban
  }
 },
 mark = {
  line = {
   color = waingoban,
  },
  label = {color = waingoban}
 }
}
```

```
\pgfgodefineplayer[from = wain-neutral]{wain-black}{
 stone = {
  fill = {
   color = wainblack
  }
 },
 move = {
  label = {
   color = waingrid
  }
 },
 mark = {
  line = {
   color = waingrid,
  },
  label = {color = waingrid}
 }
}
```

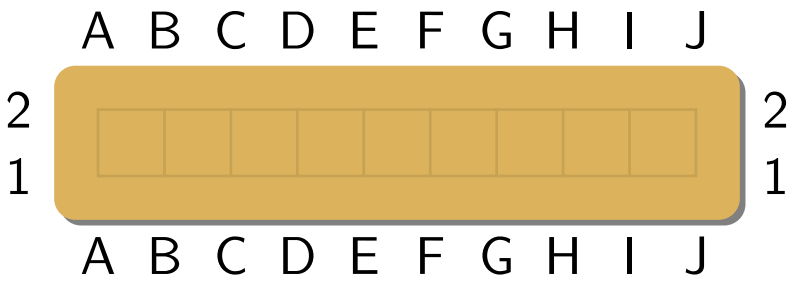
## 9.2 Boards

### 9.2.1 pgfgo



```
\pgfgodefinegoban{pgfgo}{
 background = {
  fill = {
   color = pgfgo,
   opacity = 1},
  line = {
   color = pgfgo!10!black,
   opacity = 1},
  drop shadow = {true},
  rounded corners = {
   inner = 0pt,
   outer = 2pt
  }},
 grid = {
  line = {
   color = pgfgo!10!black,
   width = 1pt}},
 label = {
  text color = black
 }
}
```

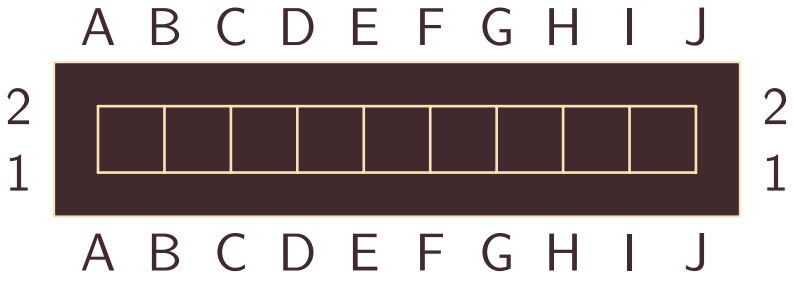
### 9.2.2 pgfgosmooth




```
\definecolor{pgfgosmooth-grid}{HTML}{C7A253}

\pgfgodefinegoban[from = pgfgo]{pgfgosmooth}{
 background = {
  line = false,
  rounded corners = {
   inner = 2pt,
   outer = 8pt
  },
  drop shadow = {true}},
 grid = {
  line = {color = pgfgosmooth-grid}
  }
}
```

### 9.2.3 wain



```
\pgfgodefinegoban[from = default]{wain}{
 background = {
  fill = {
   color = waingoban,
   opacity = 1},
  line = {
   color = waingrid,
   opacity = 1}},
 grid = {
  line = {
   color = waingrid,
   width = 1pt}
  },
 label = {text color = waingoban}
}
```